



VMware Cloud Foundation – STIG Hardening Overview

NSX Manager/Edge

9.1.0.0

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Revision History

Date	Description of Change
May 2026	Initial Release

Overview

Broadcom is a trusted partner in highly secure, mission critical systems around the world, including the United States Department of Defense (DoD). In the DoD, all IT systems must adhere to the rigorous Risk Management Framework (RMF) as defined in DoDI 8510.01. A critical component of RMF is the mandatory implementation of Security Technical Implementation Guides (STIGs) and Security Requirements Guidelines (SRGs) as maintained by the Defense Information Systems Agency (DISA).

To support our customers, NSX Manager and Edge appliances are evaluated against the following standards:

- DISA Canonical Ubuntu 24.04 LTS Security Technical Implementation Guide, Version 1 Release 5

This report will document the product's compliance with this guidance, including any deviations.

Applicability

The contents of this document are applicable to the following product versions:

- VMware Cloud Foundation 9.1.0.0

Disclaimer

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NSX Compliance

VMware NSX is a powerful network virtualization solution for VMware Cloud Foundation that enables network connectivity, operations, and scale. NSX takes a software-defined approach to networking that extends across data centers, clouds, and application frameworks. With NSX, network services are brought closer to the application wherever it's running, from virtual machines to containers to physical servers. NSX delivers cloud operational efficiency for the network independent of the underlying hardware. NSX reproduces the entire network model in software, enabling any network topology from simple to complex multitier networks to be created and provisioned in seconds.

To deliver these capabilities, the NSX Manager and Edge virtual appliances are deployed with VCF.

- NSX Edge nodes - VMs that provide network services to all the NSX components. Also known as edge transport nodes.
- NSX Managers - VMs that provide a browser-based GUI for administering the NSX environment.

Ubuntu 24.04 Compliance: Overall

The NSX Manager and Edge appliances are built on the Ubuntu 24.04 OS. The results below are based on auditing the NSX appliances against the latest Canonical 24.04 LTS STIG release.

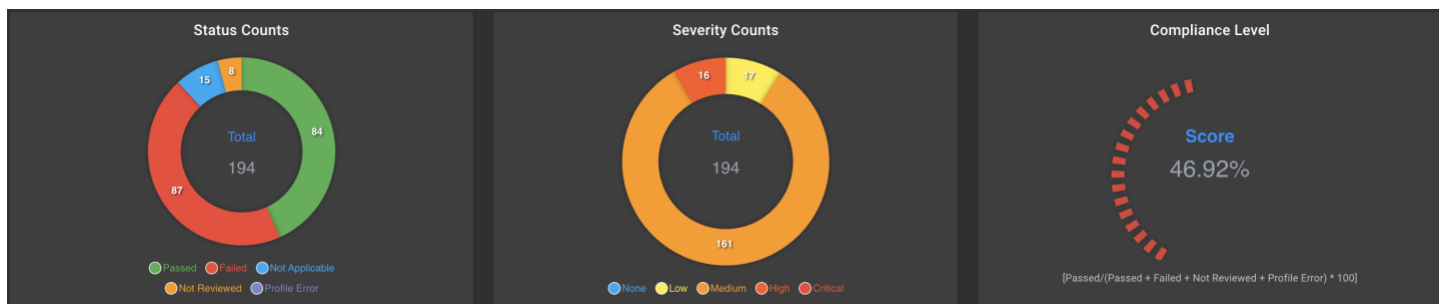


Figure 1: Overall Compliance for the NSX Manager Appliances

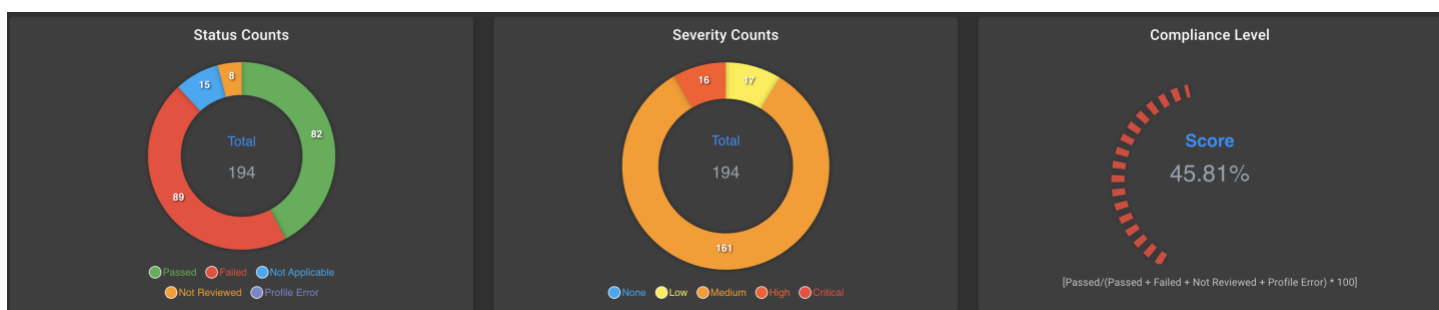


Figure 2: Overall Compliance for the NSX Edge Appliances

A full list of controls and their statuses is available in the Appendix sections of this document.

NSX Manager Appliances Ubuntu 24.04 Compliance: Exceptions

Controls listed in the exceptions table are findings. If post deployment remediation is possible it will be detailed in the justification column.

Control ID	NIST 800-53	Title	Justification
UBTU-24-100010	CM-6 b	Ubuntu 24.04 LTS must not have the "systemd-timesyncd" package installed.	NTPD is used for time synchronization instead of Chrony.
UBTU-24-100020	CM-6 b	Ubuntu 24.04 LTS must not have the "ntp" package installed.	NTPD is used for time synchronization instead of Chrony.
UBTU-24-100100	SI-6 a	Ubuntu 24.04 LTS must use a file integrity tool to verify correct operation of all security functions.	Resolution included in product roadmap.
UBTU-24-100130	SI-6 d, CM-3 (5)	Ubuntu 24.04 LTS must notify designated personnel if baseline configurations are changed in an unauthorized manner. The file integrity tool must notify the system administrator (SA) when changes to the baseline configuration or anomalies in the operation of any security functions are discovered.	Resolution included in product roadmap.
UBTU-24-100300	AC-17 (1)	Ubuntu 24.04 LTS must have an application firewall installed in order to control remote access methods.	Iptables is used instead of UFW for the local firewall solution.
UBTU-24-100450	AU-4 (1)	Ubuntu 24.04 LTS audit event multiplexor must be configured to offload audit logs onto a different system or storage media from the system being audited.	Audit logs are offloaded by configuring log forwarding for NSX. A rule exists for this configuration in the NSX Manager STIG.
UBTU-24-100600	CM-6 b	Ubuntu 24.04 LTS must have the "libpam-pwquality" package installed.	Resolution included in product roadmap. The pam_cracklib module is currently used instead of pam_pwquality.
UBTU-24-100650	IA-2 (6) (a), IA-2 (6) (b), IA-2 (1), IA-2 (2)	Ubuntu 24.04 LTS must have the "SSSD" package installed.	Configuring MFA to the local OS on VMware appliances is currently unsupported. MFA to product interface is supported via an identity provider.
UBTU-24-100660	IA-2 (6) (a), IA-2 (6) (b), IA-2 (1), IA-2 (2)	Ubuntu 24.04 LTS must use the "SSSD" package for multifactor authentication services.	Configuring MFA to the local OS on VMware appliances is currently unsupported. MFA to product interface is supported via an identity provider.
UBTU-24-100700	CM-6 b	Ubuntu 24.04 LTS must have the "chrony" package installed.	NTPD is used for time synchronization instead of Chrony.
UBTU-24-100820	AC-17 (2), MA-4 (6), SC-8 (1)	Ubuntu 24.04 LTS must configure the SSH daemon to use FIPS 140-3 approved ciphers to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.	Resolution included in product roadmap.
UBTU-24-100830	AC-17 (2), MA-4 (6), SC-8 (1)	Ubuntu 24.04 LTS must configure the SSH daemon to use Message Authentication Codes (MACs) employing FIPS 140-3 approved cryptographic hashes to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.	Resolution included in product roadmap.
UBTU-24-100840	AC-17 (2)	Ubuntu 24.04 LTS SSH server must be configured to use only FIPS 140-3 validated key exchange algorithms.	Resolution included in product roadmap.
UBTU-24-100850	AC-17 (2)	Ubuntu 24.04 LTS must configure the SSH client to use FIPS 140-3 approved ciphers to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.	Resolution included in product roadmap.

Control ID	NIST 800-53	Title	Justification
UBTU-24-100860	AC-17 (2)	Ubuntu 24.04 LTS SSH client must be configured to use only Message Authentication Codes (MACs) employing FIPS 140-3 validated cryptographic hash algorithms.	Resolution included in product roadmap.
UBTU-24-100900	IA-2 (12)	Ubuntu 24.04 LTS must accept Personal Identity Verification (PIV) credentials.	Configuring MFA to the local OS on VMware appliances is currently unsupported. MFA to product interface is supported via an identity provider.
UBTU-24-100910	IA-2 (12)	Ubuntu 24.04 LTS must accept Personal Identity Verification (PIV) credentials managed through the Privileged Access Management (PAM) framework.	Configuring MFA to the local OS on VMware appliances is currently unsupported. MFA to product interface is supported via an identity provider.
UBTU-24-101000	AC-11 (1), AC-11 a	Ubuntu 24.04 LTS must allow users to directly initiate a session lock for all connection types.	The vlock package is not installed. Session locks for the console or SSH sessions are generally unsupported on VCF appliances and instead session timeouts are implemented to mitigate sessions that are left open. Logging out is recommended instead of session locking.
UBTU-24-102010	AU-14 (1)	Ubuntu 24.04 LTS must initiate session audits at system startup.	Resolution included in product roadmap.
UBTU-24-200000	AC-10	Ubuntu 24.04 LTS must limit the number of concurrent sessions to 10 for all accounts and/or account types.	Resolution included in product roadmap.
UBTU-24-200060	AC-12	Ubuntu 24.04 LTS must automatically terminate a user session after inactivity timeouts have expired.	Resolution included in product roadmap.
UBTU-24-200260	AC-2 (3) (a), AC-2 (3) (b)	Ubuntu 24.04 LTS must disable account identifiers (individuals, groups, roles, and devices) after 35 days of inactivity.	Resolution included in product roadmap.
UBTU-24-200270	AU-12 c	Ubuntu 24.04 LTS must audit any script or executable called by cron as root or by any privileged user.	Resolution included in product roadmap.
UBTU-24-200580	AC-6 (8), AC-6 (9), MA-3 (5)	Ubuntu 24.04 LTS must prevent all software from executing at higher privilege levels than users executing the software and the audit system must be configured to audit the execution of privileged functions.	Resolution included in product roadmap.
UBTU-24-200610	AC-7 a, AC-7 b	Ubuntu 24.04 LTS must automatically lock an account until the locked account is released by an administrator when three unsuccessful logon attempts have been made.	Resolution included in product roadmap.
UBTU-24-200640	AC-8 a, AC-8 c 1, AC-8 c 2, AC-8 c 2, AC-8 c 3	Ubuntu 24.04 LTS must display the Standard Mandatory DOD Notice and Consent Banner before granting access to via an SSH logon.	The DOD login banner is not configured out of the box as it is not appropriate for all customers.
UBTU-24-300001	CM-14	Ubuntu 24.04 LTS Advance Package Tool (APT) must be configured to prevent the installation of patches, service packs, device drivers, or Ubuntu 24.04 LTS components without verification they have been digitally signed using a certificate that is recognized and approved by the organization.	Resolution included in product roadmap.
UBTU-24-300007	CM-5 (6)	Ubuntu 24.04 LTS library files must be owned by root.	Resolution included in product roadmap.
UBTU-24-300008	CM-5 (6)	Ubuntu 24.04 LTS library directories must be owned by root.	Resolution included in product roadmap.
UBTU-24-300009	CM-5 (6)	Ubuntu 24.04 LTS library files must be group-owned by root or a system account.	Resolution included in product roadmap.
UBTU-24-300010	CM-5 (6)	Ubuntu 24.04 LTS library directories must be group-owned by root.	Resolution included in product roadmap.

Control ID	NIST 800-53	Title	Justification
UBTU-24-300012	CM-5 (6)	Ubuntu 24.04 LTS must have system commands owned by root or a system account.	Resolution included in product roadmap.
UBTU-24-300013	CM-5 (6)	Ubuntu 24.04 LTS must have system commands group-owned by root or a system account.	Resolution included in product roadmap.
UBTU-24-300014	IA-5 (1) (b)	Ubuntu 24.04 LTS must prevent the use of dictionary words for passwords.	Resolution included in product roadmap.
UBTU-24-300016	CM-6 b	Ubuntu 24.04 LTS must be configured so that when passwords are changed or new passwords are established, pwquality must be used.	Resolution included in product roadmap. This requirement is currently implemented with the pam_cracklib module. The pam_cracklib module is currently used instead of pam_pwquality.
UBTU-24-300017	CM-6 b	Ubuntu 24.04 LTS must enforce a delay of at least four seconds between logon prompts following a failed logon attempt.	Resolution included in product roadmap.
UBTU-24-300020	CM-6 b	Ubuntu 24.04 LTS must require users to provide a password for privilege escalation.	Resolution included in product roadmap.
UBTU-24-300026	CM-6 b	Ubuntu 24.04 LTS must disable the x86 Ctrl-Alt-Delete key sequence.	Resolution included in product roadmap.
UBTU-24-300029	CM-6 b	Ubuntu 24.04 LTS must generate audit records for all events that affect the systemd journal files.	Resolution included in product roadmap.
UBTU-24-300030	CM-6 b	Ubuntu 24.04 LTS default filesystem permissions must be defined in such a way that all authenticated users can read and modify only their own files.	Resolution included in product roadmap.
UBTU-24-300039	CM-7 (9) (b), IA-3	Ubuntu 24.04 LTS must disable automatic mounting of Universal Serial Bus (USB) mass storage driver.	Resolution included in product roadmap.
UBTU-24-400020	IA-2 (1), IA-2 (2)	Ubuntu 24.04 LTS must implement smart card logins for multifactor authentication for local and network access to privileged and nonprivileged accounts.	Configuring MFA to the local OS on VMware appliances is currently unsupported. MFA to product interface is supported via an identity provider.
UBTU-24-400110	IA-2 (5)	Ubuntu 24.04 LTS must prevent direct login to the root account.	Resolution included in product roadmap.
UBTU-24-400260	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one uppercase character be used.	Resolution included in product roadmap. This requirement is currently implemented with the pam_cracklib module. The pam_cracklib module is currently used instead of pam_pwquality.
UBTU-24-400270	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one lowercase character be used.	Resolution included in product roadmap. This requirement is currently implemented with the pam_cracklib module. The pam_cracklib module is currently used instead of pam_pwquality.
UBTU-24-400280	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one numeric character be used.	Resolution included in product roadmap. This requirement is currently implemented with the pam_cracklib module. The pam_cracklib module is currently used instead of pam_pwquality.

Control ID	NIST 800-53	Title	Justification
UBTU-24-400290	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must require the change of at least eight characters when passwords are changed.	Resolution included in product roadmap. This is configurable post deployment but must be done via the supported methods and not directly to the underlying OS configuration. A rule exists for this configuration in the NSX Manager STIG.
UBTU-24-400300	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce 24 hours/1 day as the minimum password lifetime. Passwords for new users must have a 24 hours/1 day minimum password lifetime restriction.	Resolution included in product roadmap.
UBTU-24-400310	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce a 60-day maximum password lifetime restriction. Passwords for new users must have a 60-day maximum password lifetime restriction.	Resolution included in product roadmap.
UBTU-24-400320	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce a minimum 15-character password length.	Resolution included in product roadmap. This is configurable post deployment but must be done via the supported methods and not directly to the underlying OS configuration. A rule exists for this configuration in the NSX Manager STIG.
UBTU-24-400330	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one special character be used.	Resolution included in product roadmap.
UBTU-24-400370	IA-5 (2) (a) (2)	Ubuntu 24.04 LTS must map the authenticated identity to the user or group account for PKI-based authentication.	Configuring MFA to the local OS on VMware appliances is currently unsupported. MFA to product interface is supported via an identity provider.
UBTU-24-600000	SC-10	Ubuntu 24.04 LTS must immediately terminate all network connections associated with SSH traffic after a period of inactivity.	Resolution included in product roadmap.
UBTU-24-600010	SC-10	Ubuntu 24.04 LTS must immediately terminate all network connections associated with SSH traffic at the end of the session or after 10 minutes of inactivity.	Resolution included in product roadmap.
UBTU-24-600060	SC-23 (5)	Ubuntu 24.04 LTS must use DOD PKI-established certificate authorities (CAs) for verification of the establishment of protected sessions.	Adding DOD Root CAs to the trust store by default is not appropriate for all customers and is N/A in this context.
UBTU-24-600150	SC-4	Ubuntu 24.04 LTS must set a sticky bit on all public directories to prevent unauthorized and unintended information transferred via shared system resources.	Resolution included in product roadmap.
UBTU-24-600160	SC-45, SC-45 (1) (a)	Ubuntu 24.04 LTS must compare internal information system clocks at least every 24 hours with an authoritative time server.	NTPD is used for time synchronization instead of Chrony.
UBTU-24-600190	SC-5 (2)	Ubuntu 24.04 LTS must be configured to use TCP syncookies.	Resolution included in product roadmap.
UBTU-24-700010	SI-11 a	Ubuntu 24.04 LTS must generate error messages that provide information necessary for corrective actions without revealing information that could be exploited by adversaries.	Resolution included in product roadmap.
UBTU-24-700020	SI-11 a	Ubuntu 24.04 LTS must generate system journal entries without revealing information that could be exploited by adversaries.	Resolution included in product roadmap.
UBTU-24-700030	SI-11 a	Ubuntu 24.04 LTS must be configured so that the "journalctl" command is not accessible by unauthorized users.	Resolution included in product roadmap.

Control ID	NIST 800-53	Title	Justification
UBTU-24-700120	SI-11 b	Ubuntu 24.04 LTS must configure the /var/log directory to have mode "0755" or less permissive.	Resolution included in product roadmap.
UBTU-24-700320	SI-2 (6)	Ubuntu 24.04 LTS must be configured so that Advance Package Tool (APT) removes all software components after updated versions have been installed.	Resolution included in product roadmap.
UBTU-24-900040	AU-12 b	Ubuntu 24.04 LTS must be configured so that audit configuration files are not write-accessible by unauthorized users.	Resolution included in product roadmap.
UBTU-24-900090	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the mount command.	Resolution included in product roadmap.
UBTU-24-900100	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the umount command.	Resolution included in product roadmap.
UBTU-24-900120	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the ssh-keysign command.	Resolution included in product roadmap.
UBTU-24-900160	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the creat, open, openat, open_by_handle_at, truncate, and ftruncate system calls.	Resolution included in product roadmap.
UBTU-24-900180	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the sudoedit command.	Resolution included in product roadmap.
UBTU-24-900210	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chcon command.	Resolution included in product roadmap.
UBTU-24-900220	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the apparmor_parser command.	Resolution included in product roadmap.
UBTU-24-900230	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the setfacl command.	Resolution included in product roadmap.
UBTU-24-900240	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chacl command.	Resolution included in product roadmap.
UBTU-24-900280	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the unix_update command.	Resolution included in product roadmap.
UBTU-24-900310	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the usermod command.	Resolution included in product roadmap.
UBTU-24-900330	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the pam_timestamp_check command.	Resolution included in product roadmap.
UBTU-24-900340	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the init_module and finit_module syscalls.	Resolution included in product roadmap.
UBTU-24-900350	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the delete_module syscall.	Resolution included in product roadmap.
UBTU-24-900520	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to modify the /etc/sudoers.d directory occur.	Resolution included in product roadmap.
UBTU-24-900540	AU-12 c	Ubuntu 24.04 LTS must generate audit records for any successful/unsuccessful use of unlink, unlinkat, rename, renameat, and rmdir system calls.	Resolution included in product roadmap.
UBTU-24-900740	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to use the kmod command.	Resolution included in product roadmap.

Control ID	NIST 800-53	Title	Justification
UBTU-24-900750	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to use the fdisk command.	Resolution included in product roadmap.
UBTU-24-900950	AU-4 (1)	Ubuntu 24.04 LTS must have a crontab script running weekly to offload audit events of standalone systems.	Resolution included in product roadmap. A rule exists for this configuration in the NSX Manager STIG.
UBTU-24-900960	AU-5 (1)	Ubuntu 24.04 LTS must immediately notify the system administrator (SA) and information system security officer (ISSO) (at a minimum) when allocated audit record storage volume reaches 75 percent of the repository maximum audit record storage capacity.	Resolution included in product roadmap.
UBTU-24-901300	AU-9 a, AU-9 a	Ubuntu 24.04 LTS must be configured so that audit log files are not read or write-accessible by unauthorized users.	Resolution included in product roadmap.
UBTU-24-901350	AU-9 a, AU-9 a, AU-9 a	Ubuntu 24.04 LTS must permit only authorized groups ownership of the audit log files.	Resolution included in product roadmap.
UBTU-24-909890	AU-9 (3)	Ubuntu 24.04 LTS must use cryptographic mechanisms to protect the integrity of audit tools.	Resolution included in product roadmap.

NSX Manager Appliances Ubuntu 24.04 Compliance: Not Applicable

Controls listed in the not applicable table were determined to be not applicable in the NSX context.

Control ID	NIST 800-53	Title	Justification
UBTU-24-100110	SI-6 a	Ubuntu 24.04 LTS must configure AIDE to perform file integrity checking on the file system if installed.	AIDE not installed so this rule is not applicable.
UBTU-24-100120	SI-6 b	Ubuntu 24.04 LTS must be configured so that the script which runs each 30 days or less to check file integrity is the default one.	This is a manual review.
UBTU-24-100310	AC-17 (1)	Ubuntu 24.04 LTS must enable and run the Uncomplicated Firewall (ufw).	Iptables is used instead of UFW for the local firewall solution.
UBTU-24-200020	AC-11 a	Ubuntu 24.04 LTS must initiate a graphical session lock after 10 minutes of inactivity.	A GUI is not installed or supported on NSX manager or edge appliances.
UBTU-24-200040	AC-11 b	Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user interface automount function.	A GUI is not installed or supported on NSX manager or edge appliances.
UBTU-24-200041	IA-3, IA-3	Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user interface autorun function.	A GUI is not installed or supported on NSX manager or edge appliances.
UBTU-24-200042	AC-11 b, AC-11 a, AC-11 a	Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user smart card removal action.	A GUI is not installed or supported on NSX manager or edge appliances.
UBTU-24-200043	AC-11 (1)	Ubuntu 24.04 LTS must conceal, via the session lock, information previously visible on the display with a publicly viewable image.	A GUI is not installed or supported on NSX manager or edge appliances.
UBTU-24-200650	AC-8 a	Ubuntu 24.04 LTS must enable the graphical user logon banner to display the Standard Mandatory DOD Notice and Consent Banner before granting local access to the system via a graphical user logon.	A GUI is not installed or supported on NSX manager or edge appliances.
UBTU-24-200660	AC-8 a	Ubuntu 24.04 LTS must display the Standard Mandatory DOD Notice and Consent Banner before granting local access to the system via a graphical user logon.	A GUI is not installed or supported on NSX manager or edge appliances.

Control ID	NIST 800-53	Title	Justification
UBTU-24-200680	AC-8 b	Ubuntu 24.04 LTS must be configured to enforce the acknowledgement of the Standard Mandatory DOD Notice and Consent Banner for all SSH connections.	This is a manual review. The DOD login banner is not configured out of the box as it is not appropriate for all customers.
UBTU-24-300025	CM-6 b	Ubuntu 24.04 LTS must disable the x86 Ctrl-Alt-Delete key sequence if a graphical user interface is installed.	A GUI is not installed or supported on NSX manager or edge appliances.
UBTU-24-300041	CM-7 b	Ubuntu 24.04 LTS must be configured to prohibit or restrict the use of functions, ports, protocols, and/or services, as defined in the Ports, Protocols, and Services Management–†Category Assurance List (PPSM CAL) and vulnerability assessments.	By default, all traffic in and out is dropped by policy unless explicitly allowed. Details on specific ports and protocols used in NSX can be found at: https://ports.broadcom.com/
UBTU-24-400060	IA-2 (12)	Ubuntu 24.04 LTS must electronically verify Personal Identity Verification (PIV) credentials.	Smartcards are not supported for local logins.
UBTU-24-400340	IA-5 (13)	Ubuntu 24.04 LTS must be configured such that Pluggable Authentication Module (PAM) prohibits the use of cached authentications after one day.	Smartcards are not supported for local logins.
UBTU-24-400360	IA-5 (2) (b) (1), SC-17 b	Ubuntu 24.04 LTS, for PKI-based authentication, SSSD must validate certificates by constructing a certification path (which includes status information) to an accepted trust anchor.	Smartcards are not supported for local logins.
UBTU-24-400375	IA-5 (2) (b) (1), SC-17 b	Ubuntu 24.04 LTS, for PKI-based authentication, Privileged Access Management (PAM) must validate certificates by constructing a certification path (which includes status information) to an accepted trust anchor.	Smartcards are not supported for local logins.
UBTU-24-400380	IA-5 (2) (b) (2)	Ubuntu 24.04 LTS for PKI-based authentication, must implement a local cache of revocation data in case of the inability to access revocation information via the network.	Smartcards are not supported for local logins.
UBTU-24-600090	SC-28, SC-28 (1), SC-28 (1), SC-28 (3)	Ubuntu 24.04 LTS handling data requiring "data at rest" protections must employ cryptographic mechanisms to prevent unauthorized disclosure and modification of the information at rest.	Data at rest encryption is not implemented locally, however, the NSX virtual machines can reside on storage that provide data at rest encryption such as vSAN.
UBTU-24-600180	SC-45 (1) (b)	Ubuntu 24.04 LTS must synchronize internal information system clocks to the authoritative time source when the time difference is greater than one second.	NTPD is used for time synchronization instead of Chrony.
UBTU-24-600200	SC-5 a	Ubuntu 24.04 LTS must configure the uncomplicated firewall to rate-limit impacted network interfaces.	Iptables is used instead of UFW for the local firewall solution.
UBTU-24-600230	SC-8	Ubuntu 24.04 LTS must disable all wireless network adapters.	Wireless adapters do not exist by default on NSX manager or edge appliances.
UBTU-24-700400	CM-6 b	Ubuntu 24.04 LTS must be a vendor-supported release.	Verifying vendor support status via the Ubuntu Pro subscription is not applicable to Ubuntu based VCF appliances as Ubuntu Pro is not available on these systems. Ubuntu 24.04 LTS standard support is valid until April 2029

NSX Edge Appliances Ubuntu 24.04 Compliance: Exceptions

Controls listed in the exceptions table are findings. If post deployment remediation is possible it will be detailed in the justification column.

Control ID	NIST 800-53	Title	Justification
UBTU-24-100010	CM-6 b	Ubuntu 24.04 LTS must not have the "systemd-timesyncd" package installed.	NTPD is used for time synchronization instead of Chrony.
UBTU-24-100020	CM-6 b	Ubuntu 24.04 LTS must not have the "ntp" package installed.	NTPD is used for time synchronization instead of Chrony.
UBTU-24-100100	SI-6 a	Ubuntu 24.04 LTS must use a file integrity tool to verify correct operation of all security functions.	Resolution included in product roadmap.
UBTU-24-100130	SI-6 d, CM-3 (5)	Ubuntu 24.04 LTS must notify designated personnel if baseline configurations are changed in an unauthorized manner. The file integrity tool must notify the system administrator (SA) when changes to the baseline configuration or anomalies in the operation of any security functions are discovered.	Resolution included in product roadmap.
UBTU-24-100300	AC-17 (1)	Ubuntu 24.04 LTS must have an application firewall installed in order to control remote access methods.	Iptables is used instead of UFW for the local firewall solution.
UBTU-24-100450	AU-4 (1)	Ubuntu 24.04 LTS audit event multiplexor must be configured to offload audit logs onto a different system or storage media from the system being audited.	Audit logs are offloaded by configuring log forwarding for NSX. A rule exists for this configuration in the NSX Manager STIG.
UBTU-24-100600	CM-6 b	Ubuntu 24.04 LTS must have the "libpam-pwquality" package installed.	Resolution included in product roadmap. The pam_cracklib module is currently used instead of pam_pwquality.
UBTU-24-100650	IA-2 (6) (a), IA-2 (6) (b), IA-2 (1), IA-2 (2)	Ubuntu 24.04 LTS must have the "SSSD" package installed.	Configuring MFA to the local OS on VMware appliances is currently unsupported. MFA to product interface is supported via an identity provider.
UBTU-24-100660	IA-2 (6) (a), IA-2 (6) (b), IA-2 (1), IA-2 (2)	Ubuntu 24.04 LTS must use the "SSSD" package for multifactor authentication services.	Configuring MFA to the local OS on VMware appliances is currently unsupported. MFA to product interface is supported via an identity provider.
UBTU-24-100700	CM-6 b	Ubuntu 24.04 LTS must have the "chrony" package installed.	NTPD is used for time synchronization instead of Chrony.
UBTU-24-100820	AC-17 (2), MA-4 (6), SC-8 (1)	Ubuntu 24.04 LTS must configure the SSH daemon to use FIPS 140-3 approved ciphers to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.	Resolution included in product roadmap.
UBTU-24-100830	AC-17 (2), MA-4 (6), SC-8 (1)	Ubuntu 24.04 LTS must configure the SSH daemon to use Message Authentication Codes (MACs) employing FIPS 140-3 approved cryptographic hashes to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.	Resolution included in product roadmap.
UBTU-24-100840	AC-17 (2)	Ubuntu 24.04 LTS SSH server must be configured to use only FIPS 140-3 validated key exchange algorithms.	Resolution included in product roadmap.
UBTU-24-100850	AC-17 (2)	Ubuntu 24.04 LTS must configure the SSH client to use FIPS 140-3 approved ciphers to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.	Resolution included in product roadmap.

Control ID	NIST 800-53	Title	Justification
UBTU-24-100860	AC-17 (2)	Ubuntu 24.04 LTS SSH client must be configured to use only Message Authentication Codes (MACs) employing FIPS 140-3 validated cryptographic hash algorithms.	Resolution included in product roadmap.
UBTU-24-100900	IA-2 (12)	Ubuntu 24.04 LTS must accept Personal Identity Verification (PIV) credentials.	Configuring MFA to the local OS on VMware appliances is currently unsupported. MFA to product interface is supported via an identity provider.
UBTU-24-100910	IA-2 (12)	Ubuntu 24.04 LTS must accept Personal Identity Verification (PIV) credentials managed through the Privileged Access Management (PAM) framework.	Configuring MFA to the local OS on VMware appliances is currently unsupported. MFA to product interface is supported via an identity provider.
UBTU-24-101000	AC-11 (1), AC-11 a	Ubuntu 24.04 LTS must allow users to directly initiate a session lock for all connection types.	The vlock package is not installed. Session locks for the console or SSH sessions are generally unsupported on VCF appliances and instead session timeouts are implemented to mitigate sessions that are left open. Logging out is recommended instead of session locking.
UBTU-24-102010	AU-14 (1)	Ubuntu 24.04 LTS must initiate session audits at system startup.	Resolution included in product roadmap.
UBTU-24-200000	AC-10	Ubuntu 24.04 LTS must limit the number of concurrent sessions to 10 for all accounts and/or account types.	Resolution included in product roadmap.
UBTU-24-200060	AC-12	Ubuntu 24.04 LTS must automatically terminate a user session after inactivity timeouts have expired.	Resolution included in product roadmap.
UBTU-24-200260	AC-2 (3) (a), AC-2 (3) (b)	Ubuntu 24.04 LTS must disable account identifiers (individuals, groups, roles, and devices) after 35 days of inactivity.	Resolution included in product roadmap.
UBTU-24-200270	AU-12 c	Ubuntu 24.04 LTS must audit any script or executable called by cron as root or by any privileged user.	Resolution included in product roadmap.
UBTU-24-200580	AC-6 (8), AC-6 (9), MA-3 (5)	Ubuntu 24.04 LTS must prevent all software from executing at higher privilege levels than users executing the software and the audit system must be configured to audit the execution of privileged functions.	Resolution included in product roadmap.
UBTU-24-200610	AC-7 a, AC-7 b	Ubuntu 24.04 LTS must automatically lock an account until the locked account is released by an administrator when three unsuccessful logon attempts have been made.	Resolution included in product roadmap.
UBTU-24-200640	AC-8 a, AC-8 c 1, AC-8 c 2, AC-8 c 2, AC-8 c 3	Ubuntu 24.04 LTS must display the Standard Mandatory DOD Notice and Consent Banner before granting access to via an SSH logon.	The DOD login banner is not configured out of the box as it is not appropriate for all customers.
UBTU-24-300001	CM-14	Ubuntu 24.04 LTS Advance Package Tool (APT) must be configured to prevent the installation of patches, service packs, device drivers, or Ubuntu 24.04 LTS components without verification they have been digitally signed using a certificate that is recognized and approved by the organization.	Resolution included in product roadmap.
UBTU-24-300007	CM-5 (6)	Ubuntu 24.04 LTS library files must be owned by root.	Resolution included in product roadmap.
UBTU-24-300008	CM-5 (6)	Ubuntu 24.04 LTS library directories must be owned by root.	Resolution included in product roadmap.
UBTU-24-300009	CM-5 (6)	Ubuntu 24.04 LTS library files must be group-owned by root or a system account.	Resolution included in product roadmap.
UBTU-24-300010	CM-5 (6)	Ubuntu 24.04 LTS library directories must be group-owned by root.	Resolution included in product roadmap.

Control ID	NIST 800-53	Title	Justification
UBTU-24-300012	CM-5 (6)	Ubuntu 24.04 LTS must have system commands owned by root or a system account.	Resolution included in product roadmap.
UBTU-24-300013	CM-5 (6)	Ubuntu 24.04 LTS must have system commands group-owned by root or a system account.	Resolution included in product roadmap.
UBTU-24-300014	IA-5 (1) (b)	Ubuntu 24.04 LTS must prevent the use of dictionary words for passwords.	Resolution included in product roadmap.
UBTU-24-300016	CM-6 b	Ubuntu 24.04 LTS must be configured so that when passwords are changed or new passwords are established, pwquality must be used.	Resolution included in product roadmap. This requirement is currently implemented with the pam_cracklib module. The pam_cracklib module is currently used instead of pam_pwquality.
UBTU-24-300017	CM-6 b	Ubuntu 24.04 LTS must enforce a delay of at least four seconds between logon prompts following a failed logon attempt.	Resolution included in product roadmap.
UBTU-24-300020	CM-6 b	Ubuntu 24.04 LTS must require users to provide a password for privilege escalation.	Resolution included in product roadmap.
UBTU-24-300026	CM-6 b	Ubuntu 24.04 LTS must disable the x86 Ctrl-Alt-Delete key sequence.	Resolution included in product roadmap.
UBTU-24-300029	CM-6 b	Ubuntu 24.04 LTS must generate audit records for all events that affect the systemd journal files.	Resolution included in product roadmap.
UBTU-24-300030	CM-6 b	Ubuntu 24.04 LTS default filesystem permissions must be defined in such a way that all authenticated users can read and modify only their own files.	Resolution included in product roadmap.
UBTU-24-300039	CM-7 (9) (b), IA-3	Ubuntu 24.04 LTS must disable automatic mounting of Universal Serial Bus (USB) mass storage driver.	Resolution included in product roadmap.
UBTU-24-400020	IA-2 (1), IA-2 (2)	Ubuntu 24.04 LTS must implement smart card logins for multifactor authentication for local and network access to privileged and nonprivileged accounts.	Configuring MFA to the local OS on VMware appliances is currently unsupported. MFA to product interface is supported via an identity provider.
UBTU-24-400110	IA-2 (5)	Ubuntu 24.04 LTS must prevent direct login to the root account.	Resolution included in product roadmap.
UBTU-24-400260	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one uppercase character be used.	Resolution included in product roadmap. This requirement is currently implemented with the pam_cracklib module. The pam_cracklib module is currently used instead of pam_pwquality.
UBTU-24-400270	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one lowercase character be used.	Resolution included in product roadmap. This requirement is currently implemented with the pam_cracklib module. The pam_cracklib module is currently used instead of pam_pwquality.
UBTU-24-400280	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one numeric character be used.	Resolution included in product roadmap. This requirement is currently implemented with the pam_cracklib module. The pam_cracklib module is currently used instead of pam_pwquality.

Control ID	NIST 800-53	Title	Justification
UBTU-24-400290	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must require the change of at least eight characters when passwords are changed.	Resolution included in product roadmap. This is configurable post deployment but must be done via the supported methods and not directly to the underlying OS configuration. A rule exists for this configuration in the NSX Manager STIG.
UBTU-24-400300	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce 24 hours/1 day as the minimum password lifetime. Passwords for new users must have a 24 hours/1 day minimum password lifetime restriction.	Resolution included in product roadmap.
UBTU-24-400310	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce a 60-day maximum password lifetime restriction. Passwords for new users must have a 60-day maximum password lifetime restriction.	Resolution included in product roadmap.
UBTU-24-400320	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce a minimum 15-character password length.	Resolution included in product roadmap. This is configurable post deployment but must be done via the supported methods and not directly to the underlying OS configuration. A rule exists for this configuration in the NSX Manager STIG.
UBTU-24-400330	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one special character be used.	Resolution included in product roadmap.
UBTU-24-400370	IA-5 (2) (a) (2)	Ubuntu 24.04 LTS must map the authenticated identity to the user or group account for PKI-based authentication.	Configuring MFA to the local OS on VMware appliances is currently unsupported. MFA to product interface is supported via an identity provider.
UBTU-24-600000	SC-10	Ubuntu 24.04 LTS must immediately terminate all network connections associated with SSH traffic after a period of inactivity.	Resolution included in product roadmap.
UBTU-24-600010	SC-10	Ubuntu 24.04 LTS must immediately terminate all network connections associated with SSH traffic at the end of the session or after 10 minutes of inactivity.	Resolution included in product roadmap.
UBTU-24-600030	SC-13 b	Ubuntu 24.04 LTS must implement NIST FIPS-validated cryptography to protect classified information and for the following: To provision digital signatures, to generate cryptographic hashes, and to protect unclassified information requiring confidentiality and cryptographic protection in accordance with applicable federal laws, Executive Orders, directives, policies, regulations, and standards.	Resolution included in product roadmap.
UBTU-24-600060	SC-23 (5)	Ubuntu 24.04 LTS must use DOD PKI-established certificate authorities (CAs) for verification of the establishment of protected sessions.	Adding DOD Root CAs to the trust store by default is not appropriate for all customers and is N/A in this context.
UBTU-24-600070	SC-24	Ubuntu 24.04 LTS must disable kernel core dumps.	Resolution included in product roadmap.
UBTU-24-600160	SC-45, SC-45 (1) (a)	Ubuntu 24.04 LTS must compare internal information system clocks at least every 24 hours with an authoritative time server.	NTPD is used for time synchronization instead of Chrony.
UBTU-24-600190	SC-5 (2)	Ubuntu 24.04 LTS must be configured to use TCP syncookies.	Resolution included in product roadmap.
UBTU-24-700010	SI-11 a	Ubuntu 24.04 LTS must generate error messages that provide information necessary for corrective actions without revealing information that could be exploited by adversaries.	Resolution included in product roadmap.
UBTU-24-700020	SI-11 a	Ubuntu 24.04 LTS must generate system journal entries without revealing information that could be exploited by adversaries.	Resolution included in product roadmap.

Control ID	NIST 800-53	Title	Justification
UBTU-24-700030	SI-11 a	Ubuntu 24.04 LTS must be configured so that the "journalctl" command is not accessible by unauthorized users.	Resolution included in product roadmap.
UBTU-24-700070	SI-11 b	Ubuntu 24.04 LTS must configure the files used by the system journal to be group-owned by "systemd-journal".	Resolution included in product roadmap.
UBTU-24-700120	SI-11 b	Ubuntu 24.04 LTS must configure the /var/log directory to have mode "0755" or less permissive.	Resolution included in product roadmap.
UBTU-24-700320	SI-2 (6)	Ubuntu 24.04 LTS must be configured so that Advance Package Tool (APT) removes all software components after updated versions have been installed.	Resolution included in product roadmap.
UBTU-24-900040	AU-12 b	Ubuntu 24.04 LTS must be configured so that audit configuration files are not write-accessible by unauthorized users.	Resolution included in product roadmap.
UBTU-24-900090	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the mount command.	Resolution included in product roadmap.
UBTU-24-900100	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the umount command.	Resolution included in product roadmap.
UBTU-24-900120	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the ssh-keysign command.	Resolution included in product roadmap.
UBTU-24-900160	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the creat, open, openat, open_by_handle_at, truncate, and ftruncate system calls.	Resolution included in product roadmap.
UBTU-24-900180	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the sudoedit command.	Resolution included in product roadmap.
UBTU-24-900210	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chcon command.	Resolution included in product roadmap.
UBTU-24-900220	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the apparmor_parser command.	Resolution included in product roadmap.
UBTU-24-900230	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the setfacl command.	Resolution included in product roadmap.
UBTU-24-900240	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chacl command.	Resolution included in product roadmap.
UBTU-24-900280	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the unix_update command.	Resolution included in product roadmap.
UBTU-24-900310	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the usermod command.	Resolution included in product roadmap.
UBTU-24-900330	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the pam_timestamp_check command.	Resolution included in product roadmap.
UBTU-24-900340	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the init_module and finit_module syscalls.	Resolution included in product roadmap.
UBTU-24-900350	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the delete_module syscall.	Resolution included in product roadmap.
UBTU-24-900520	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to modify the /etc/sudoers.d directory occur.	Resolution included in product roadmap.

Control ID	NIST 800-53	Title	Justification
UBTU-24-900540	AU-12 c	Ubuntu 24.04 LTS must generate audit records for any successful/unsuccessful use of unlink, unlinkat, rename, renameat, and rmdir system calls.	Resolution included in product roadmap.
UBTU-24-900740	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to use the kmod command.	Resolution included in product roadmap.
UBTU-24-900750	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to use the fdisk command.	Resolution included in product roadmap.
UBTU-24-900950	AU-4 (1)	Ubuntu 24.04 LTS must have a crontab script running weekly to offload audit events of standalone systems.	Resolution included in product roadmap. A rule exists for this configuration in the NSX Manager STIG.
UBTU-24-900960	AU-5 (1)	Ubuntu 24.04 LTS must immediately notify the system administrator (SA) and information system security officer (ISSO) (at a minimum) when allocated audit record storage volume reaches 75 percent of the repository maximum audit record storage capacity.	Resolution included in product roadmap.
UBTU-24-901300	AU-9 a, AU-9 a	Ubuntu 24.04 LTS must be configured so that audit log files are not read or write-accessible by unauthorized users.	Resolution included in product roadmap.
UBTU-24-901350	AU-9 a, AU-9 a, AU-9 a	Ubuntu 24.04 LTS must permit only authorized groups ownership of the audit log files.	Resolution included in product roadmap.
UBTU-24-909890	AU-9 (3)	Ubuntu 24.04 LTS must use cryptographic mechanisms to protect the integrity of audit tools.	Resolution included in product roadmap.

NSX Edge Appliances Ubuntu 24.04 Compliance: Not Applicable

Controls listed in the not applicable table were determined to be not applicable in the NSX context.

Control ID	NIST 800-53	Title	Justification
UBTU-24-100110	SI-6 a	Ubuntu 24.04 LTS must configure AIDE to perform file integrity checking on the file system if installed.	AIDE not installed so this rule is not applicable.
UBTU-24-100120	SI-6 b	Ubuntu 24.04 LTS must be configured so that the script which runs each 30 days or less to check file integrity is the default one.	This is a manual review.
UBTU-24-100310	AC-17 (1)	Ubuntu 24.04 LTS must enable and run the Uncomplicated Firewall (ufw).	Iptables is used instead of UFW for the local firewall solution.
UBTU-24-200020	AC-11 a	Ubuntu 24.04 LTS must initiate a graphical session lock after 10 minutes of inactivity.	A GUI is not installed or supported on NSX manager or edge appliances.
UBTU-24-200040	AC-11 b	Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user interface automount function.	A GUI is not installed or supported on NSX manager or edge appliances.
UBTU-24-200041	IA-3, IA-3	Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user interface autorun function.	A GUI is not installed or supported on NSX manager or edge appliances.
UBTU-24-200042	AC-11 b, AC-11 a, AC-11 a	Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user smart card removal action.	A GUI is not installed or supported on NSX manager or edge appliances.
UBTU-24-200043	AC-11 (1)	Ubuntu 24.04 LTS must conceal, via the session lock, information previously visible on the display with a publicly viewable image.	A GUI is not installed or supported on NSX manager or edge appliances.

Control ID	NIST 800-53	Title	Justification
UBTU-24-200650	AC-8 a	Ubuntu 24.04 LTS must enable the graphical user logon banner to display the Standard Mandatory DOD Notice and Consent Banner before granting local access to the system via a graphical user logon.	A GUI is not installed or supported on NSX manager or edge appliances.
UBTU-24-200660	AC-8 a	Ubuntu 24.04 LTS must display the Standard Mandatory DOD Notice and Consent Banner before granting local access to the system via a graphical user logon.	A GUI is not installed or supported on NSX manager or edge appliances.
UBTU-24-200680	AC-8 b	Ubuntu 24.04 LTS must be configured to enforce the acknowledgement of the Standard Mandatory DOD Notice and Consent Banner for all SSH connections.	This is a manual review. The DOD login banner is not configured out of the box as it is not appropriate for all customers.
UBTU-24-300025	CM-6 b	Ubuntu 24.04 LTS must disable the x86 Ctrl-Alt-Delete key sequence if a graphical user interface is installed.	A GUI is not installed or supported on NSX manager or edge appliances.
UBTU-24-300041	CM-7 b	Ubuntu 24.04 LTS must be configured to prohibit or restrict the use of functions, ports, protocols, and/or services, as defined in the Ports, Protocols, and Services Management Category Assurance List (PPSM CAL) and vulnerability assessments.	By default, all traffic in and out is dropped by policy unless explicitly allowed. Details on specific ports and protocols used in NSX can be found at: https://ports.broadcom.com/
UBTU-24-400060	IA-2 (12)	Ubuntu 24.04 LTS must electronically verify Personal Identity Verification (PIV) credentials.	Smartcards are not supported for local logins.
UBTU-24-400340	IA-5 (13)	Ubuntu 24.04 LTS must be configured such that Pluggable Authentication Module (PAM) prohibits the use of cached authentications after one day.	Smartcards are not supported for local logins.
UBTU-24-400360	IA-5 (2) (b) (1), SC-17 b	Ubuntu 24.04 LTS, for PKI-based authentication, SSSD must validate certificates by constructing a certification path (which includes status information) to an accepted trust anchor.	Smartcards are not supported for local logins.
UBTU-24-400375	IA-5 (2) (b) (1), SC-17 b	Ubuntu 24.04 LTS, for PKI-based authentication, Privileged Access Management (PAM) must validate certificates by constructing a certification path (which includes status information) to an accepted trust anchor.	Smartcards are not supported for local logins.
UBTU-24-400380	IA-5 (2) (b) (2)	Ubuntu 24.04 LTS for PKI-based authentication, must implement a local cache of revocation data in case of the inability to access revocation information via the network.	Smartcards are not supported for local logins.
UBTU-24-600090	SC-28, SC-28 (1), SC-28 (1), SC-28 (3)	Ubuntu 24.04 LTS handling data requiring "data at rest" protections must employ cryptographic mechanisms to prevent unauthorized disclosure and modification of the information at rest.	Data at rest encryption is not implemented locally, however, the NSX virtual machines can reside on storage that provide data at rest encryption such as vSAN.
UBTU-24-600180	SC-45 (1) (b)	Ubuntu 24.04 LTS must synchronize internal information system clocks to the authoritative time source when the time difference is greater than one second.	NTPD is used for time synchronization instead of Chrony.
UBTU-24-600200	SC-5 a	Ubuntu 24.04 LTS must configure the uncomplicated firewall to rate-limit impacted network interfaces.	Iptables is used instead of UFW for the local firewall solution.
UBTU-24-600230	SC-8	Ubuntu 24.04 LTS must disable all wireless network adapters.	Wireless adapters do not exist by default on NSX manager or edge appliances.
UBTU-24-700400	CM-6 b	Ubuntu 24.04 LTS must be a vendor-supported release.	Verifying vendor support status via the Ubuntu Pro subscription is not applicable to Ubuntu based VCF appliances as Ubuntu Pro is not available on these systems. Ubuntu 24.04 LTS standard support is valid until April 2029

Frequently Asked Questions

Can customers make changes to the NSX Manager and Edge appliances?

No it is not supported to remediate the findings listed in this document at this time.

Where can I find the Canonical Ubuntu 24.04 LTS STIG?

The Canonical Ubuntu 24.04 LTS STIG may be found at:

<https://public.cyber.mil/stigs/>

What is a STIG Readiness Guide?

More information about STIG Readiness Guides can be found at:

<https://www.vmware.com/docs/vmw-stig-program-overview>

What does the “status” column in the control list tables mean?

Status Definitions	
Passed	The compliance check passed.
Failed	The compliance check failed.
Not Applicable	The control was determined to be N/A in this context.
Not Reviewed	These controls were skipped as the conditions of the test did not exist on the system or require manual review and count as failures unless otherwise attested to manually.

Appendix: Full Control List – NSX Manager Appliances

Control ID	NIST 800-53	Title	Status
UBTU-24-100010	CM-6 b	Ubuntu 24.04 LTS must not have the "systemd-timesyncd" package installed.	Failed
UBTU-24-100020	CM-6 b	Ubuntu 24.04 LTS must not have the "ntp" package installed.	Failed
UBTU-24-100030	IA-5 (1) (c)	Ubuntu 24.04 LTS must not have the telnet package installed.	Passed
UBTU-24-100040	CM-7 a	Ubuntu 24.04 LTS must not have the rsh-server package installed.	Passed
UBTU-24-100050	CM-7 a	Ubuntu 24.04 LTS must not have the nfs-kernel-server package installed.	Passed
UBTU-24-100100	SI-6 a	Ubuntu 24.04 LTS must use a file integrity tool to verify correct operation of all security functions.	Failed
UBTU-24-100110	SI-6 a	Ubuntu 24.04 LTS must configure AIDE to perform file integrity checking on the file system if installed.	Not Applicable
UBTU-24-100120	SI-6 b	Ubuntu 24.04 LTS must be configured so that the script which runs each 30 days or less to check file integrity is the default one.	Not Reviewed
UBTU-24-100130	SI-6 d, CM-3 (5)	Ubuntu 24.04 LTS must notify designated personnel if baseline configurations are changed in an unauthorized manner. The file integrity tool must notify the system administrator (SA) when changes to the baseline configuration or anomalies in the operation of any security functions are discovered.	Failed
UBTU-24-100200	SC-24	Ubuntu 24.04 LTS must be configured to preserve log records from failure events.	Passed
UBTU-24-100300	AC-17 (1)	Ubuntu 24.04 LTS must have an application firewall installed in order to control remote access methods.	Failed
UBTU-24-100310	AC-17 (1)	Ubuntu 24.04 LTS must enable and run the Uncomplicated Firewall (ufw).	Not Reviewed
UBTU-24-100400	CM-5 (1) (b), AU-12 (3), AU-12 a, AU-12 c, AU-3 (1), AU-3 a, AU-3 b, AU-3 c, AU-3 d, AU-3 e, AU-6 (4), AU-7 (1), AU-7 a, AU-7 a, AU-7 a, AU-7 a, AU-7 a, AU-7 b, AU-7 b	Ubuntu 24.04 LTS must have the "auditd" package installed.	Passed
UBTU-24-100410	CM-5 (1) (b), AU-12 (3), AU-12 a, AU-12 c, AU-3 (1), AU-3 a, AU-3 b, AU-3 c, AU-3 d, AU-3 e, AU-6 (4), AU-7 (1), AU-7 a, AU-7 a, AU-7 a, AU-7 a, AU-7 a, AU-7 b, AU-7 b	Ubuntu 24.04 LTS must produce audit records and reports containing information to establish when, where, what type, the source, and the outcome for all DOD-defined auditable events and actions in near real time.	Passed

Control ID	NIST 800-53	Title	Status
UBTU-24-100450	AU-4 (1)	Ubuntu 24.04 LTS audit event multiplexor must be configured to offload audit logs onto a different system or storage media from the system being audited.	Failed
UBTU-24-100500	AC-3 (4), CM-7 (2), CM-7 (5) (b)	Ubuntu 24.04 LTS must have AppArmor installed.	Passed
UBTU-24-100510	CM-7 (2), AC-6 (10), CM-7 (5) (b)	Ubuntu 24.04 LTS must be configured to use AppArmor.	Passed
UBTU-24-100600	CM-6 b	Ubuntu 24.04 LTS must have the "libpam-pwquality" package installed.	Failed
UBTU-24-100650	IA-2 (6) (a), IA-2 (6) (b), IA-2 (1), IA-2 (2)	Ubuntu 24.04 LTS must have the "SSSD" package installed.	Failed
UBTU-24-100660	IA-2 (6) (a), IA-2 (6) (b), IA-2 (1), IA-2 (2)	Ubuntu 24.04 LTS must use the "SSSD" package for multifactor authentication services.	Failed
UBTU-24-100700	CM-6 b	Ubuntu 24.04 LTS must have the "chrony" package installed.	Failed
UBTU-24-100800	SC-8, SC-8 (2), SC-8 (2)	Ubuntu 24.04 LTS must have SSH installed.	Passed
UBTU-24-100810	SC-8, SC-8 (2), SC-8 (2)	Ubuntu 24.04 LTS must use SSH to protect the confidentiality and integrity of transmitted information.	Passed
UBTU-24-100820	AC-17 (2), MA-4 (6), SC-8 (1)	Ubuntu 24.04 LTS must configure the SSH daemon to use FIPS 140-3 approved ciphers to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.	Failed
UBTU-24-100830	AC-17 (2), MA-4 (6), SC-8 (1)	Ubuntu 24.04 LTS must configure the SSH daemon to use Message Authentication Codes (MACs) employing FIPS 140-3 approved cryptographic hashes to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.	Failed
UBTU-24-100840	AC-17 (2)	Ubuntu 24.04 LTS SSH server must be configured to use only FIPS 140-3 validated key exchange algorithms.	Failed
UBTU-24-100850	AC-17 (2)	Ubuntu 24.04 LTS must configure the SSH client to use FIPS 140-3 approved ciphers to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.	Failed
UBTU-24-100860	AC-17 (2)	Ubuntu 24.04 LTS SSH client must be configured to use only Message Authentication Codes (MACs) employing FIPS 140-3 validated cryptographic hash algorithms.	Failed
UBTU-24-100900	IA-2 (12)	Ubuntu 24.04 LTS must accept Personal Identity Verification (PIV) credentials.	Failed
UBTU-24-100910	IA-2 (12)	Ubuntu 24.04 LTS must accept Personal Identity Verification (PIV) credentials managed through the Privileged Access Management (PAM) –†framework.	Failed
UBTU-24-101000	AC-11 (1), AC-11 a	Ubuntu 24.04 LTS must allow users to directly initiate a session lock for all connection types.	Failed
UBTU-24-102000	AC-3	Ubuntu 24.04 LTS when booted must require authentication upon booting into single-user and maintenance modes.	Passed
UBTU-24-102010	AU-14 (1)	Ubuntu 24.04 LTS must initiate session audits at system startup.	Failed
UBTU-24-200000	AC-10	Ubuntu 24.04 LTS must limit the number of concurrent sessions to 10 for all accounts and/or account types.	Failed
UBTU-24-200020	AC-11 a	Ubuntu 24.04 LTS must initiate a graphical session lock after 10 minutes of inactivity.	Not Applicable

Control ID	NIST 800-53	Title	Status
UBTU-24-200040	AC-11 b	Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user interface automount function.	Not Applicable
UBTU-24-200041	IA-3, IA-3	Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user interface autorun function.	Not Applicable
UBTU-24-200042	AC-11 b, AC-11 a, AC-11 a	Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user smart card removal action.	Not Applicable
UBTU-24-200043	AC-11 (1)	Ubuntu 24.04 LTS must conceal, via the session lock, information previously visible on the display with a publicly viewable image.	Not Applicable
UBTU-24-200060	AC-12	Ubuntu 24.04 LTS must automatically terminate a user session after inactivity timeouts have expired.	Failed
UBTU-24-200090	AC-17 (1)	Ubuntu 24.04 LTS must monitor remote access methods.	Passed
UBTU-24-200250	AC-2 (2), AC-2 (2)	Ubuntu 24.04 LTS must automatically remove or disable emergency accounts after 72 hours.	Passed
UBTU-24-200260	AC-2 (3) (a), AC-2 (3) (b)	Ubuntu 24.04 LTS must disable account identifiers (individuals, groups, roles, and devices) after 35 days of inactivity.	Failed
UBTU-24-200270	AU-12 c	Ubuntu 24.04 LTS must audit any script or executable called by cron as root or by any privileged user.	Failed
UBTU-24-200280	AC-2 (4), AC-2 (4), AC-2 (4), AC-2 (4), AU-12 c	Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/passwd.	Passed
UBTU-24-200290	AC-2 (4), AC-2 (4), AC-2 (4), AC-2 (4), AU-12 c	Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/group.	Passed
UBTU-24-200300	AC-2 (4), AC-2 (4), AC-2 (4), AC-2 (4), AU-12 c	Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/shadow.	Passed
UBTU-24-200310	AC-2 (4), AC-2 (4), AC-2 (4), AC-2 (4), AU-12 c	Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/gshadow.	Passed
UBTU-24-200320	AC-2 (4), AC-2 (4), AC-2 (4), AC-2 (4), AU-12 c	Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/opasswd.	Passed
UBTU-24-200580	AC-6 (8), AC-6 (9), MA-3 (5)	Ubuntu 24.04 LTS must prevent all software from executing at higher privilege levels than users executing the software and the audit system must be configured to audit the execution of privileged functions.	Failed
UBTU-24-200610	AC-7 a, AC-7 b	Ubuntu 24.04 LTS must automatically lock an account until the locked account is released by an administrator when three unsuccessful logon attempts have been made.	Failed
UBTU-24-200640	AC-8 a, AC-8 c 1, AC-8 c 2, AC-8 c 2, AC-8 c 3	Ubuntu 24.04 LTS must display the Standard Mandatory DOD Notice and Consent Banner before granting access to via an SSH logon.	Failed
UBTU-24-200650	AC-8 a	Ubuntu 24.04 LTS must enable the graphical user logon banner to display the Standard Mandatory DOD Notice and Consent Banner before granting local access to the system via a graphical user logon.	Not Applicable

Control ID	NIST 800-53	Title	Status
UBTU-24-200660	AC-8 a	Ubuntu 24.04 LTS must display the Standard Mandatory DOD Notice and Consent Banner before granting local access to the system via a graphical user logon.	Not Applicable
UBTU-24-200680	AC-8 b	Ubuntu 24.04 LTS must be configured to enforce the acknowledgement of the Standard Mandatory DOD Notice and Consent Banner for all SSH connections.	Not Reviewed
UBTU-24-300001	CM-14	Ubuntu 24.04 LTS Advance Package Tool (APT) must be configured to prevent the installation of patches, service packs, device drivers, or Ubuntu 24.04 LTS components without verification they have been digitally signed using a certificate that is recognized and approved by the organization.	Failed
UBTU-24-300006	CM-5 (6)	Ubuntu 24.04 LTS library files must have mode 0755 or less permissive.	Passed
UBTU-24-300007	CM-5 (6)	Ubuntu 24.04 LTS library files must be owned by root.	Failed
UBTU-24-300008	CM-5 (6)	Ubuntu 24.04 LTS library directories must be owned by root.	Failed
UBTU-24-300009	CM-5 (6)	Ubuntu 24.04 LTS library files must be group-owned by root or a system account.	Failed
UBTU-24-300010	CM-5 (6)	Ubuntu 24.04 LTS library directories must be group-owned by root.	Failed
UBTU-24-300011	CM-5 (6)	Ubuntu 24.04 LTS must have system commands set to a mode of 0755 or less permissive.	Passed
UBTU-24-300012	CM-5 (6)	Ubuntu 24.04 LTS must have system commands owned by root or a system account.	Failed
UBTU-24-300013	CM-5 (6)	Ubuntu 24.04 LTS must have system commands group-owned by root or a system account.	Failed
UBTU-24-300014	IA-5 (1) (b)	Ubuntu 24.04 LTS must prevent the use of dictionary words for passwords.	Failed
UBTU-24-300016	CM-6 b	Ubuntu 24.04 LTS must be configured so that when passwords are changed or new passwords are established, pwquality must be used.	Failed
UBTU-24-300017	CM-6 b	Ubuntu 24.04 LTS must enforce a delay of at least four seconds between logon prompts following a failed logon attempt.	Failed
UBTU-24-300019	IA-11, SC-11 b	Ubuntu 24.04 LTS must restrict privilege elevation to authorized personnel.	Passed
UBTU-24-300020	CM-6 b	Ubuntu 24.04 LTS must require users to provide a password for privilege escalation.	Failed
UBTU-24-300021	CM-6 b	Ubuntu 24.04 LTS must require users to reauthenticate for privilege escalation or when changing roles.	Passed
UBTU-24-300022	CM-6 b	Ubuntu 24.04 LTS must be configured so that remote X connections are disabled, unless to fulfill documented and validated mission requirements.	Passed
UBTU-24-300023	CM-6 b	Ubuntu 24.04 LTS SSH daemon must prevent remote hosts from connecting to the proxy display.	Passed
UBTU-24-300025	CM-6 b	Ubuntu 24.04 LTS must disable the x86 Ctrl-Alt-Delete key sequence if a graphical user interface is installed.	Not Applicable
UBTU-24-300026	CM-6 b	Ubuntu 24.04 LTS must disable the x86 Ctrl-Alt-Delete key sequence.	Failed
UBTU-24-300027	CM-6 b	Ubuntu 24.04 LTS must not have accounts configured with blank or null passwords.	Passed
UBTU-24-300028	CM-6 b	Ubuntu 24.04 LTS must not allow accounts configured in Pluggable Authentication Modules (PAM) with blank or null passwords.	Passed
UBTU-24-300029	CM-6 b	Ubuntu 24.04 LTS must generate audit records for all events that affect the systemd journal files.	Failed

Control ID	NIST 800-53	Title	Status
UBTU-24-300030	CM-6 b	Ubuntu 24.04 LTS default filesystem permissions must be defined in such a way that all authenticated users can read and modify only their own files.	Failed
UBTU-24-300031	CM-6 b	Ubuntu 24.04 LTS must not allow unattended or automatic login via SSH.	Passed
UBTU-24-300039	CM-7 (9) (b), IA-3	Ubuntu 24.04 LTS must disable automatic mounting of Universal Serial Bus (USB) mass storage driver.	Failed
UBTU-24-300041	CM-7 b	Ubuntu 24.04 LTS must be configured to prohibit or restrict the use of functions, ports, protocols, and/or services, as defined in the Ports, Protocols, and Services Management–†Category Assurance List (PPSM CAL) and vulnerability assessments.	Not Reviewed
UBTU-24-400000	IA-2, IA-8	Ubuntu 24.04 LTS must uniquely identify interactive users.	Passed
UBTU-24-400020	IA-2 (1), IA-2 (2)	Ubuntu 24.04 LTS must implement smart card logins for multifactor authentication for local and network access to privileged and nonprivileged accounts.	Failed
UBTU-24-400030	IA-2 (1), IA-2 (2)	Ubuntu 24.04 LTS must implement smart card logins for multifactor authentication for local and network access to privileged and nonprivileged accounts over SSH.	Passed
UBTU-24-400060	IA-2 (12)	Ubuntu 24.04 LTS must electronically verify Personal Identity Verification (PIV) credentials.	Not Applicable
UBTU-24-400110	IA-2 (5)	Ubuntu 24.04 LTS must prevent direct login to the root account.	Failed
UBTU-24-400220	IA-5 (1) (d)	Ubuntu 24.04 LTS must store only encrypted representations of passwords.	Passed
UBTU-24-400260	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one uppercase character be used.	Failed
UBTU-24-400270	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one lowercase character be used.	Failed
UBTU-24-400280	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one numeric character be used.	Failed
UBTU-24-400290	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must require the change of at least eight characters when passwords are changed.	Failed
UBTU-24-400300	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce 24 hours/1 day as the minimum password lifetime. Passwords for new users must have a 24 hours/1 day minimum password lifetime restriction.	Failed
UBTU-24-400310	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce a 60-day maximum password lifetime restriction. Passwords for new users must have a 60-day maximum password lifetime restriction.	Failed
UBTU-24-400320	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce a minimum 15-character password length.	Failed
UBTU-24-400330	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one special character be used.	Failed
UBTU-24-400340	IA-5 (13)	Ubuntu 24.04 LTS must be configured such that Pluggable Authentication Module (PAM) prohibits the use of cached authentications after one day.	Not Applicable
UBTU-24-400360	IA-5 (2) (b) (1), SC-17 b	Ubuntu 24.04 LTS, for PKI-based authentication, SSSD must validate certificates by constructing a certification path (which includes status information) to an accepted trust anchor.	Not Applicable
UBTU-24-400370	IA-5 (2) (a) (2)	Ubuntu 24.04 LTS must map the authenticated identity to the user or group account for PKI-based authentication.	Failed
UBTU-24-400375	IA-5 (2) (b) (1), SC-17 b	Ubuntu 24.04 LTS, for PKI-based authentication, Privileged Access Management (PAM) must validate certificates by constructing a certification path (which includes status information) to an accepted trust anchor.	Not Applicable

Control ID	NIST 800-53	Title	Status
UBTU-24-400380	IA-5 (2) (b) (2)	Ubuntu 24.04 LTS for PKI-based authentication, must implement a local cache of revocation data in case of the inability to access revocation information via the network.	Not Applicable
UBTU-24-400400	IA-7	Ubuntu 24.04 LTS must encrypt all stored passwords with a FIPS 140-3 approved cryptographic hashing algorithm.	Passed
UBTU-24-500010	MA-4 (1) (a), AU-12 c	Ubuntu 24.04 LTS must generate audit records for privileged activities, nonlocal maintenance, diagnostic sessions, and other system-level access.	Passed
UBTU-24-500050	MA-4 c	Ubuntu 24.04 LTS must use strong authenticators in establishing nonlocal maintenance and diagnostic sessions.	Passed
UBTU-24-600000	SC-10	Ubuntu 24.04 LTS must immediately terminate all network connections associated with SSH traffic after a period of inactivity.	Failed
UBTU-24-600010	SC-10	Ubuntu 24.04 LTS must immediately terminate all network connections associated with SSH traffic at the end of the session or after 10 minutes of inactivity.	Failed
UBTU-24-600030	SC-13 b	Ubuntu 24.04 LTS must implement NIST FIPS-validated cryptography to protect classified information and for the following: To provision digital signatures, to generate cryptographic hashes, and to protect unclassified information requiring confidentiality and cryptographic protection in accordance with applicable federal laws, Executive Orders, directives, policies, regulations, and standards.	Passed
UBTU-24-600060	SC-23 (5)	Ubuntu 24.04 LTS must use DOD PKI-established certificate authorities (CAs) for verification of the establishment of protected sessions.	Failed
UBTU-24-600070	SC-24	Ubuntu 24.04 LTS must disable kernel core dumps.	Passed
UBTU-24-600090	SC-28, SC-28 (1), SC-28 (1), SC-28 (3)	Ubuntu 24.04 LTS handling data requiring "data at rest" protections must employ cryptographic mechanisms to prevent unauthorized disclosure and modification of the information at rest.	Not Reviewed
UBTU-24-600130	SC-3	Ubuntu 24.04 LTS must ensure only users who need access to security functions are part of sudo group.	Passed
UBTU-24-600140	SC-4	Ubuntu 24.04 LTS must restrict access to the kernel message buffer.	Passed
UBTU-24-600150	SC-4	Ubuntu 24.04 LTS must set a sticky bit on all public directories to prevent unauthorized and unintended information transferred via shared system resources.	Failed
UBTU-24-600160	SC-45, SC-45 (1) (a)	Ubuntu 24.04 LTS must compare internal information system clocks at least every 24 hours with an authoritative time server.	Failed
UBTU-24-600180	SC-45 (1) (b)	Ubuntu 24.04 LTS must synchronize internal information system clocks to the authoritative time source when the time difference is greater than one second.	Not Reviewed
UBTU-24-600190	SC-5 (2)	Ubuntu 24.04 LTS must be configured to use TCP syncookies.	Failed
UBTU-24-600200	SC-5 a	Ubuntu 24.04 LTS must configure the uncomplicated firewall to rate-limit impacted network interfaces.	Not Reviewed
UBTU-24-600230	SC-8	Ubuntu 24.04 LTS must disable all wireless network adapters.	Not Applicable
UBTU-24-700010	SI-11 a	Ubuntu 24.04 LTS must generate error messages that provide information necessary for corrective actions without revealing information that could be exploited by adversaries.	Failed
UBTU-24-700020	SI-11 a	Ubuntu 24.04 LTS must generate system journal entries without revealing information that could be exploited by adversaries.	Failed
UBTU-24-700030	SI-11 a	Ubuntu 24.04 LTS must be configured so that the "journalctl" command is not accessible by unauthorized users.	Failed
UBTU-24-700040	SI-11 b	Ubuntu 24.04 LTS must be configured so that the "journalctl" command is owned by "root".	Passed

Control ID	NIST 800-53	Title	Status
UBTU-24-700050	SI-11 b	Ubuntu 24.04 LTS must be configured so that the "journalctl" command is group-owned by "root".	Passed
UBTU-24-700060	SI-11 b	Ubuntu 24.04 LTS must configure the directories used by the system journal to be group-owned by "systemd-journal".	Passed
UBTU-24-700070	SI-11 b	Ubuntu 24.04 LTS must configure the files used by the system journal to be group-owned by "systemd-journal".	Passed
UBTU-24-700080	SI-11 b	Ubuntu 24.04 LTS must configure the directories used by the system journal to be owned by "root".	Passed
UBTU-24-700090	SI-11 b	Ubuntu 24.04 LTS must configure the files used by the system journal to be owned by "root".	Passed
UBTU-24-700100	SI-11 b	Ubuntu 24.04 LTS must configure the /var/log directory to be group-owned by syslog.	Passed
UBTU-24-700110	SI-11 b	Ubuntu 24.04 LTS must configure the /var/log directory to be owned by root.	Passed
UBTU-24-700120	SI-11 b	Ubuntu 24.04 LTS must configure the /var/log directory to have mode "0755" or less permissive.	Failed
UBTU-24-700130	SI-11 b	Ubuntu 24.04 LTS must configure the /var/log/syslog file to be group-owned by adm.	Passed
UBTU-24-700140	SI-11 b	Ubuntu 24.04 LTS must configure /var/log/syslog file to be owned by syslog.	Passed
UBTU-24-700150	SI-11 b	Ubuntu 24.04 LTS must configure /var/log/syslog file with mode "0640" or less permissive.	Passed
UBTU-24-700300	SI-16	Ubuntu 24.04 LTS must implement nonexecutable data to protect its memory from unauthorized code execution.	Passed
UBTU-24-700310	SI-16	Ubuntu 24.04 LTS must implement address space layout randomization to protect its memory from unauthorized code execution.	Passed
UBTU-24-700320	SI-2 (6)	Ubuntu 24.04 LTS must be configured so that Advance Package Tool (APT) removes all software components after updated versions have been installed.	Failed
UBTU-24-700400	CM-6 b	Ubuntu 24.04 LTS must be a vendor-supported release.	Not Reviewed
UBTU-24-900040	AU-12 b	Ubuntu 24.04 LTS must be configured so that audit configuration files are not write-accessible by unauthorized users.	Failed
UBTU-24-900050	AU-12 b	Ubuntu 24.04 LTS must permit only authorized accounts to own the audit configuration files.	Passed
UBTU-24-900060	AU-12 b	Ubuntu 24.04 LTS must permit only authorized groups to own the audit configuration files.	Passed
UBTU-24-900070	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the su command.	Passed
UBTU-24-900080	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chfn command.	Passed
UBTU-24-900090	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the mount command.	Failed
UBTU-24-900100	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the umount command.	Failed
UBTU-24-900110	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the ssh-agent command.	Passed
UBTU-24-900120	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the ssh-keygen command.	Failed

Control ID	NIST 800-53	Title	Status
UBTU-24-900130	AU-12 c	Ubuntu 24.04 LTS must generate audit records for any use of the setxattr, fsetxattr, lsetxattr, removexattr, fremovexattr, and lremovexattr system calls.	Passed
UBTU-24-900140	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chown, fchown, fchownat, and lchown system calls.	Passed
UBTU-24-900150	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chmod, fchmod, and fchmodat system calls.	Passed
UBTU-24-900160	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the creat, open, openat, open_by_handle_at, truncate, and ftruncate system calls.	Failed
UBTU-24-900170	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the sudo command.	Passed
UBTU-24-900180	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the sudoedit command.	Failed
UBTU-24-900190	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chsh command.	Passed
UBTU-24-900200	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the newgrp command.	Passed
UBTU-24-900210	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chcon command.	Failed
UBTU-24-900220	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the apparmor_parser command.	Failed
UBTU-24-900230	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the setfacl command.	Failed
UBTU-24-900240	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chacl command.	Failed
UBTU-24-900250	AU-12 c	Ubuntu 24.04 LTS must generate audit records for the use and modification of faillog file.	Passed
UBTU-24-900260	AU-12 c	Ubuntu 24.04 LTS must generate audit records for the use and modification of the lastlog file.	Passed
UBTU-24-900270	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the passwd command.	Passed
UBTU-24-900280	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the unix_update command.	Failed
UBTU-24-900290	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the gpasswd command.	Passed
UBTU-24-900300	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chage command.	Passed
UBTU-24-900310	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the usermod command.	Failed
UBTU-24-900320	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the crontab command.	Passed
UBTU-24-900330	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the pam_timestamp_check command.	Failed
UBTU-24-900340	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the init_module and finit_module syscalls.	Failed
UBTU-24-900350	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the delete_module syscall.	Failed
UBTU-24-900510	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to modify the /etc/sudoers file occur.	Passed

Control ID	NIST 800-53	Title	Status
UBTU-24-900520	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to modify the /etc/sudoers.d directory occur.	Failed
UBTU-24-900540	AU-12 c	Ubuntu 24.04 LTS must generate audit records for any successful/unsuccessful use of unlink, unlinkat, rename, renameat, and rmdir system calls.	Failed
UBTU-24-900590	AU-12 c	Ubuntu 24.04 LTS must generate audit records for the /var/log/wtmp file.	Passed
UBTU-24-900600	AU-12 c	Ubuntu 24.04 LTS must generate audit records for the /var/run/utmp file.	Passed
UBTU-24-900610	AU-12 c	Ubuntu 24.04 LTS must generate audit records for the /var/log/btmp file.	Passed
UBTU-24-900730	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to use modprobe command.	Passed
UBTU-24-900740	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to use the kmod command.	Failed
UBTU-24-900750	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to use the fdisk command.	Failed
UBTU-24-900920	AU-4	Ubuntu 24.04 LTS must allocate audit record storage capacity to store at least one week's worth of audit records, when audit records are not immediately sent to a central audit record storage facility.	Passed
UBTU-24-900950	AU-4 (1)	Ubuntu 24.04 LTS must have a crontab script running weekly to offload audit events of standalone systems.	Failed
UBTU-24-900960	AU-5 (1)	Ubuntu 24.04 LTS must immediately notify the system administrator (SA) and information system security officer (ISSO) (at a minimum) when allocated audit record storage volume reaches 75 percent of the repository maximum audit record storage capacity.	Failed
UBTU-24-900980	AU-5 a	Ubuntu 24.04 LTS must alert the system administrator (SA) and information system security officer (ISSO) (at a minimum) in the event of an audit processing failure.	Passed
UBTU-24-901220	AU-8 b	Ubuntu 24.04 LTS must record time stamps for audit records that can be mapped to Coordinated Universal Time (UTC) or Greenwich Mean Time (GMT).	Passed
UBTU-24-901230	AU-9, AU-9 a	Ubuntu 24.04 LTS must configure audit tools with a mode of "0755" or less permissive.	Passed
UBTU-24-901240	AU-9, AU-9 a	Ubuntu 24.04 LTS must configure audit tools to be owned by root.	Passed
UBTU-24-901250	AU-9, AU-9 a	Ubuntu 24.04 LTS must configure the audit tools to be group owned by root.	Passed
UBTU-24-901260	AU-9	Ubuntu 24.04 LTS must have directories that contain system commands set to a mode of "0755" or less permissive.	Passed
UBTU-24-901270	AU-9	Ubuntu 24.04 LTS must have directories that contain system commands owned by root.	Passed
UBTU-24-901280	AU-9	Ubuntu 24.04 LTS must have directories that contain system commands group-owned by root.	Passed
UBTU-24-901300	AU-9 a, AU-9 a	Ubuntu 24.04 LTS must be configured so that audit log files are not read or write-accessible by unauthorized users.	Failed
UBTU-24-901310	AU-9 a, AU-9 a, AU-9 a	Ubuntu 24.04 LTS must be configured to permit only authorized users ownership of the audit log files.	Passed
UBTU-24-901350	AU-9 a, AU-9 a, AU-9 a	Ubuntu 24.04 LTS must permit only authorized groups ownership of the audit log files.	Failed
UBTU-24-901380	AU-9 a	Ubuntu 24.04 LTS must be configured so that the audit log directory is not write-accessible by unauthorized users.	Passed

Control ID	NIST 800-53	Title	Status
UBTU-24-909000	AU-9 a	Ubuntu 24.04 LTS audit system must protect auditing rules from unauthorized change.	Passed
UBTU-24-909890	AU-9 (3)	Ubuntu 24.04 LTS must use cryptographic mechanisms to protect the integrity of audit tools.	Failed

Appendix: Full Control List – NSX Edge Appliances

Control ID	NIST 800-53	Title	Status
UBTU-24-100010	CM-6 b	Ubuntu 24.04 LTS must not have the "systemd-timesyncd" package installed.	Failed
UBTU-24-100020	CM-6 b	Ubuntu 24.04 LTS must not have the "ntp" package installed.	Failed
UBTU-24-100030	IA-5 (1) (c)	Ubuntu 24.04 LTS must not have the telnet package installed.	Passed
UBTU-24-100040	CM-7 a	Ubuntu 24.04 LTS must not have the rsh-server package installed.	Passed
UBTU-24-100050	CM-7 a	Ubuntu 24.04 LTS must not have the nfs-kernel-server package installed.	Passed
UBTU-24-100100	SI-6 a	Ubuntu 24.04 LTS must use a file integrity tool to verify correct operation of all security functions.	Failed
UBTU-24-100110	SI-6 a	Ubuntu 24.04 LTS must configure AIDE to perform file integrity checking on the file system if installed.	Not Applicable
UBTU-24-100120	SI-6 b	Ubuntu 24.04 LTS must be configured so that the script which runs each 30 days or less to check file integrity is the default one.	Not Reviewed
UBTU-24-100130	SI-6 d, CM-3 (5)	Ubuntu 24.04 LTS must notify designated personnel if baseline configurations are changed in an unauthorized manner. The file integrity tool must notify the system administrator (SA) when changes to the baseline configuration or anomalies in the operation of any security functions are discovered.	Failed
UBTU-24-100200	SC-24	Ubuntu 24.04 LTS must be configured to preserve log records from failure events.	Passed
UBTU-24-100300	AC-17 (1)	Ubuntu 24.04 LTS must have an application firewall installed in order to control remote access methods.	Failed
UBTU-24-100310	AC-17 (1)	Ubuntu 24.04 LTS must enable and run the Uncomplicated Firewall (ufw).	Not Reviewed
UBTU-24-100400	CM-5 (1) (b), AU-12 (3), AU-12 a, AU-12 c, AU-3 (1), AU-3 a, AU-3 b, AU-3 c, AU-3 d, AU-3 e, AU-6 (4), AU-7 (1), AU-7 a, AU-7 a, AU-7 a, AU-7 a, AU-7 a, AU-7 b, AU-7 b	Ubuntu 24.04 LTS must have the "auditd" package installed.	Passed
UBTU-24-100410	CM-5 (1) (b), AU-12 (3), AU-12 a, AU-12 c, AU-3 (1), AU-3 a, AU-3 b, AU-3 c, AU-3 d, AU-3 e, AU-6 (4), AU-7 (1), AU-7 a, AU-7 a, AU-7 a, AU-7 a, AU-7 a, AU-7 b, AU-7 b	Ubuntu 24.04 LTS must produce audit records and reports containing information to establish when, where, what type, the source, and the outcome for all DOD-defined auditable events and actions in near real time.	Passed

Control ID	NIST 800-53	Title	Status
UBTU-24-100450	AU-4 (1)	Ubuntu 24.04 LTS audit event multiplexor must be configured to offload audit logs onto a different system or storage media from the system being audited.	Failed
UBTU-24-100500	AC-3 (4), CM-7 (2), CM-7 (5) (b)	Ubuntu 24.04 LTS must have AppArmor installed.	Passed
UBTU-24-100510	CM-7 (2), AC-6 (10), CM-7 (5) (b)	Ubuntu 24.04 LTS must be configured to use AppArmor.	Passed
UBTU-24-100600	CM-6 b	Ubuntu 24.04 LTS must have the "libpam-pwquality" package installed.	Failed
UBTU-24-100650	IA-2 (6) (a), IA-2 (6) (b), IA-2 (1), IA-2 (2)	Ubuntu 24.04 LTS must have the "SSSD" package installed.	Failed
UBTU-24-100660	IA-2 (6) (a), IA-2 (6) (b), IA-2 (1), IA-2 (2)	Ubuntu 24.04 LTS must use the "SSSD" package for multifactor authentication services.	Failed
UBTU-24-100700	CM-6 b	Ubuntu 24.04 LTS must have the "chrony" package installed.	Failed
UBTU-24-100800	SC-8, SC-8 (2), SC-8 (2)	Ubuntu 24.04 LTS must have SSH installed.	Passed
UBTU-24-100810	SC-8, SC-8 (2), SC-8 (2)	Ubuntu 24.04 LTS must use SSH to protect the confidentiality and integrity of transmitted information.	Passed
UBTU-24-100820	AC-17 (2), MA-4 (6), SC-8 (1)	Ubuntu 24.04 LTS must configure the SSH daemon to use FIPS 140-3 approved ciphers to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.	Failed
UBTU-24-100830	AC-17 (2), MA-4 (6), SC-8 (1)	Ubuntu 24.04 LTS must configure the SSH daemon to use Message Authentication Codes (MACs) employing FIPS 140-3 approved cryptographic hashes to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.	Failed
UBTU-24-100840	AC-17 (2)	Ubuntu 24.04 LTS SSH server must be configured to use only FIPS 140-3 validated key exchange algorithms.	Failed
UBTU-24-100850	AC-17 (2)	Ubuntu 24.04 LTS must configure the SSH client to use FIPS 140-3 approved ciphers to prevent the unauthorized disclosure of information and/or detect changes to information during transmission.	Failed
UBTU-24-100860	AC-17 (2)	Ubuntu 24.04 LTS SSH client must be configured to use only Message Authentication Codes (MACs) employing FIPS 140-3 validated cryptographic hash algorithms.	Failed
UBTU-24-100900	IA-2 (12)	Ubuntu 24.04 LTS must accept Personal Identity Verification (PIV) credentials.	Failed
UBTU-24-100910	IA-2 (12)	Ubuntu 24.04 LTS must accept Personal Identity Verification (PIV) credentials managed through the Privileged Access Management (PAM) framework.	Failed
UBTU-24-101000	AC-11 (1), AC-11 a	Ubuntu 24.04 LTS must allow users to directly initiate a session lock for all connection types.	Failed
UBTU-24-102000	AC-3	Ubuntu 24.04 LTS when booted must require authentication upon booting into single-user and maintenance modes.	Passed
UBTU-24-102010	AU-14 (1)	Ubuntu 24.04 LTS must initiate session audits at system startup.	Failed
UBTU-24-200000	AC-10	Ubuntu 24.04 LTS must limit the number of concurrent sessions to 10 for all accounts and/or account types.	Failed
UBTU-24-200020	AC-11 a	Ubuntu 24.04 LTS must initiate a graphical session lock after 10 minutes of inactivity.	Not Applicable

Control ID	NIST 800-53	Title	Status
UBTU-24-200040	AC-11 b	Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user interface automount function.	Not Applicable
UBTU-24-200041	IA-3, IA-3	Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user interface autorun function.	Not Applicable
UBTU-24-200042	AC-11 b, AC-11 a, AC-11 a	Ubuntu 24.04 LTS must prevent a user from overriding the disabling of the graphical user smart card removal action.	Not Applicable
UBTU-24-200043	AC-11 (1)	Ubuntu 24.04 LTS must conceal, via the session lock, information previously visible on the display with a publicly viewable image.	Not Applicable
UBTU-24-200060	AC-12	Ubuntu 24.04 LTS must automatically terminate a user session after inactivity timeouts have expired.	Failed
UBTU-24-200090	AC-17 (1)	Ubuntu 24.04 LTS must monitor remote access methods.	Passed
UBTU-24-200250	AC-2 (2), AC-2 (2)	Ubuntu 24.04 LTS must automatically remove or disable emergency accounts after 72 hours.	Passed
UBTU-24-200260	AC-2 (3) (a), AC-2 (3) (b)	Ubuntu 24.04 LTS must disable account identifiers (individuals, groups, roles, and devices) after 35 days of inactivity.	Failed
UBTU-24-200270	AU-12 c	Ubuntu 24.04 LTS must audit any script or executable called by cron as root or by any privileged user.	Failed
UBTU-24-200280	AC-2 (4), AC-2 (4), AC-2 (4), AC-2 (4), AU-12 c	Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/passwd.	Passed
UBTU-24-200290	AC-2 (4), AC-2 (4), AC-2 (4), AC-2 (4), AU-12 c	Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/group.	Passed
UBTU-24-200300	AC-2 (4), AC-2 (4), AC-2 (4), AC-2 (4), AU-12 c	Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/shadow.	Passed
UBTU-24-200310	AC-2 (4), AC-2 (4), AC-2 (4), AC-2 (4), AU-12 c	Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/gshadow.	Passed
UBTU-24-200320	AC-2 (4), AC-2 (4), AC-2 (4), AC-2 (4), AU-12 c	Ubuntu 24.04 LTS must generate audit records for all account creations, modifications, disabling, and termination events that affect /etc/opasswd.	Passed
UBTU-24-200580	AC-6 (8), AC-6 (9), MA-3 (5)	Ubuntu 24.04 LTS must prevent all software from executing at higher privilege levels than users executing the software and the audit system must be configured to audit the execution of privileged functions.	Failed
UBTU-24-200610	AC-7 a, AC-7 b	Ubuntu 24.04 LTS must automatically lock an account until the locked account is released by an administrator when three unsuccessful logon attempts have been made.	Failed
UBTU-24-200640	AC-8 a, AC-8 c 1, AC-8 c 2, AC-8 c 2, AC-8 c 3	Ubuntu 24.04 LTS must display the Standard Mandatory DOD Notice and Consent Banner before granting access to via an SSH logon.	Failed
UBTU-24-200650	AC-8 a	Ubuntu 24.04 LTS must enable the graphical user logon banner to display the Standard Mandatory DOD Notice and Consent Banner before granting local access to the system via a graphical user logon.	Not Applicable

Control ID	NIST 800-53	Title	Status
UBTU-24-200660	AC-8 a	Ubuntu 24.04 LTS must display the Standard Mandatory DOD Notice and Consent Banner before granting local access to the system via a graphical user logon.	Not Applicable
UBTU-24-200680	AC-8 b	Ubuntu 24.04 LTS must be configured to enforce the acknowledgement of the Standard Mandatory DOD Notice and Consent Banner for all SSH connections.	Not Reviewed
UBTU-24-300001	CM-14	Ubuntu 24.04 LTS Advance Package Tool (APT) must be configured to prevent the installation of patches, service packs, device drivers, or Ubuntu 24.04 LTS components without verification they have been digitally signed using a certificate that is recognized and approved by the organization.	Failed
UBTU-24-300006	CM-5 (6)	Ubuntu 24.04 LTS library files must have mode 0755 or less permissive.	Passed
UBTU-24-300007	CM-5 (6)	Ubuntu 24.04 LTS library files must be owned by root.	Failed
UBTU-24-300008	CM-5 (6)	Ubuntu 24.04 LTS library directories must be owned by root.	Failed
UBTU-24-300009	CM-5 (6)	Ubuntu 24.04 LTS library files must be group-owned by root or a system account.	Failed
UBTU-24-300010	CM-5 (6)	Ubuntu 24.04 LTS library directories must be group-owned by root.	Failed
UBTU-24-300011	CM-5 (6)	Ubuntu 24.04 LTS must have system commands set to a mode of 0755 or less permissive.	Passed
UBTU-24-300012	CM-5 (6)	Ubuntu 24.04 LTS must have system commands owned by root or a system account.	Failed
UBTU-24-300013	CM-5 (6)	Ubuntu 24.04 LTS must have system commands group-owned by root or a system account.	Failed
UBTU-24-300014	IA-5 (1) (b)	Ubuntu 24.04 LTS must prevent the use of dictionary words for passwords.	Failed
UBTU-24-300016	CM-6 b	Ubuntu 24.04 LTS must be configured so that when passwords are changed or new passwords are established, pwquality must be used.	Failed
UBTU-24-300017	CM-6 b	Ubuntu 24.04 LTS must enforce a delay of at least four seconds between logon prompts following a failed logon attempt.	Failed
UBTU-24-300019	IA-11, SC-11 b	Ubuntu 24.04 LTS must restrict privilege elevation to authorized personnel.	Passed
UBTU-24-300020	CM-6 b	Ubuntu 24.04 LTS must require users to provide a password for privilege escalation.	Failed
UBTU-24-300021	CM-6 b	Ubuntu 24.04 LTS must require users to reauthenticate for privilege escalation or when changing roles.	Passed
UBTU-24-300022	CM-6 b	Ubuntu 24.04 LTS must be configured so that remote X connections are disabled, unless to fulfill documented and validated mission requirements.	Passed
UBTU-24-300023	CM-6 b	Ubuntu 24.04 LTS SSH daemon must prevent remote hosts from connecting to the proxy display.	Passed
UBTU-24-300025	CM-6 b	Ubuntu 24.04 LTS must disable the x86 Ctrl-Alt-Delete key sequence if a graphical user interface is installed.	Not Applicable
UBTU-24-300026	CM-6 b	Ubuntu 24.04 LTS must disable the x86 Ctrl-Alt-Delete key sequence.	Failed
UBTU-24-300027	CM-6 b	Ubuntu 24.04 LTS must not have accounts configured with blank or null passwords.	Passed
UBTU-24-300028	CM-6 b	Ubuntu 24.04 LTS must not allow accounts configured in Pluggable Authentication Modules (PAM) with blank or null passwords.	Passed
UBTU-24-300029	CM-6 b	Ubuntu 24.04 LTS must generate audit records for all events that affect the systemd journal files.	Failed

Control ID	NIST 800-53	Title	Status
UBTU-24-300030	CM-6 b	Ubuntu 24.04 LTS default filesystem permissions must be defined in such a way that all authenticated users can read and modify only their own files.	Failed
UBTU-24-300031	CM-6 b	Ubuntu 24.04 LTS must not allow unattended or automatic login via SSH.	Passed
UBTU-24-300039	CM-7 (9) (b), IA-3	Ubuntu 24.04 LTS must disable automatic mounting of Universal Serial Bus (USB) mass storage driver.	Failed
UBTU-24-300041	CM-7 b	Ubuntu 24.04 LTS must be configured to prohibit or restrict the use of functions, ports, protocols, and/or services, as defined in the Ports, Protocols, and Services Management–†Category Assurance List (PPSM CAL) and vulnerability assessments.	Not Reviewed
UBTU-24-400000	IA-2, IA-8	Ubuntu 24.04 LTS must uniquely identify interactive users.	Passed
UBTU-24-400020	IA-2 (1), IA-2 (2)	Ubuntu 24.04 LTS must implement smart card logins for multifactor authentication for local and network access to privileged and nonprivileged accounts.	Failed
UBTU-24-400030	IA-2 (1), IA-2 (2)	Ubuntu 24.04 LTS must implement smart card logins for multifactor authentication for local and network access to privileged and nonprivileged accounts over SSH.	Passed
UBTU-24-400060	IA-2 (12)	Ubuntu 24.04 LTS must electronically verify Personal Identity Verification (PIV) credentials.	Not Applicable
UBTU-24-400110	IA-2 (5)	Ubuntu 24.04 LTS must prevent direct login to the root account.	Failed
UBTU-24-400220	IA-5 (1) (d)	Ubuntu 24.04 LTS must store only encrypted representations of passwords.	Passed
UBTU-24-400260	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one uppercase character be used.	Failed
UBTU-24-400270	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one lowercase character be used.	Failed
UBTU-24-400280	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one numeric character be used.	Failed
UBTU-24-400290	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must require the change of at least eight characters when passwords are changed.	Failed
UBTU-24-400300	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce 24 hours/1 day as the minimum password lifetime. Passwords for new users must have a 24 hours/1 day minimum password lifetime restriction.	Failed
UBTU-24-400310	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce a 60-day maximum password lifetime restriction. Passwords for new users must have a 60-day maximum password lifetime restriction.	Failed
UBTU-24-400320	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce a minimum 15-character password length.	Failed
UBTU-24-400330	IA-5 (1) (g), IA-5 (1) (h)	Ubuntu 24.04 LTS must enforce password complexity by requiring that at least one special character be used.	Failed
UBTU-24-400340	IA-5 (13)	Ubuntu 24.04 LTS must be configured such that Pluggable Authentication Module (PAM) prohibits the use of cached authentications after one day.	Not Applicable
UBTU-24-400360	IA-5 (2) (b) (1), SC-17 b	Ubuntu 24.04 LTS, for PKI-based authentication, SSSD must validate certificates by constructing a certification path (which includes status information) to an accepted trust anchor.	Not Applicable
UBTU-24-400370	IA-5 (2) (a) (2)	Ubuntu 24.04 LTS must map the authenticated identity to the user or group account for PKI-based authentication.	Failed
UBTU-24-400375	IA-5 (2) (b) (1), SC-17 b	Ubuntu 24.04 LTS, for PKI-based authentication, Privileged Access Management (PAM) must validate certificates by constructing a certification path (which includes status information) to an accepted trust anchor.	Not Applicable

Control ID	NIST 800-53	Title	Status
UBTU-24-400380	IA-5 (2) (b) (2)	Ubuntu 24.04 LTS for PKI-based authentication, must implement a local cache of revocation data in case of the inability to access revocation information via the network.	Not Applicable
UBTU-24-400400	IA-7	Ubuntu 24.04 LTS must encrypt all stored passwords with a FIPS 140-3 approved cryptographic hashing algorithm.	Passed
UBTU-24-500010	MA-4 (1) (a), AU-12 c	Ubuntu 24.04 LTS must generate audit records for privileged activities, nonlocal maintenance, diagnostic sessions, and other system-level access.	Passed
UBTU-24-500050	MA-4 c	Ubuntu 24.04 LTS must use strong authenticators in establishing nonlocal maintenance and diagnostic sessions.	Passed
UBTU-24-600000	SC-10	Ubuntu 24.04 LTS must immediately terminate all network connections associated with SSH traffic after a period of inactivity.	Failed
UBTU-24-600010	SC-10	Ubuntu 24.04 LTS must immediately terminate all network connections associated with SSH traffic at the end of the session or after 10 minutes of inactivity.	Failed
UBTU-24-600030	SC-13 b	Ubuntu 24.04 LTS must implement NIST FIPS-validated cryptography to protect classified information and for the following: To provision digital signatures, to generate cryptographic hashes, and to protect unclassified information requiring confidentiality and cryptographic protection in accordance with applicable federal laws, Executive Orders, directives, policies, regulations, and standards.	Failed
UBTU-24-600060	SC-23 (5)	Ubuntu 24.04 LTS must use DOD PKI-established certificate authorities (CAs) for verification of the establishment of protected sessions.	Failed
UBTU-24-600070	SC-24	Ubuntu 24.04 LTS must disable kernel core dumps.	Failed
UBTU-24-600090	SC-28, SC-28 (1), SC-28 (1), SC-28 (3)	Ubuntu 24.04 LTS handling data requiring "data at rest" protections must employ cryptographic mechanisms to prevent unauthorized disclosure and modification of the information at rest.	Not Reviewed
UBTU-24-600130	SC-3	Ubuntu 24.04 LTS must ensure only users who need access to security functions are part of sudo group.	Passed
UBTU-24-600140	SC-4	Ubuntu 24.04 LTS must restrict access to the kernel message buffer.	Passed
UBTU-24-600150	SC-4	Ubuntu 24.04 LTS must set a sticky bit on all public directories to prevent unauthorized and unintended information transferred via shared system resources.	Passed
UBTU-24-600160	SC-45, SC-45 (1) (a)	Ubuntu 24.04 LTS must compare internal information system clocks at least every 24 hours with an authoritative time server.	Failed
UBTU-24-600180	SC-45 (1) (b)	Ubuntu 24.04 LTS must synchronize internal information system clocks to the authoritative time source when the time difference is greater than one second.	Not Reviewed
UBTU-24-600190	SC-5 (2)	Ubuntu 24.04 LTS must be configured to use TCP syncookies.	Failed
UBTU-24-600200	SC-5 a	Ubuntu 24.04 LTS must configure the uncomplicated firewall to rate-limit impacted network interfaces.	Not Reviewed
UBTU-24-600230	SC-8	Ubuntu 24.04 LTS must disable all wireless network adapters.	Not Applicable
UBTU-24-700010	SI-11 a	Ubuntu 24.04 LTS must generate error messages that provide information necessary for corrective actions without revealing information that could be exploited by adversaries.	Failed
UBTU-24-700020	SI-11 a	Ubuntu 24.04 LTS must generate system journal entries without revealing information that could be exploited by adversaries.	Failed
UBTU-24-700030	SI-11 a	Ubuntu 24.04 LTS must be configured so that the "journalctl" command is not accessible by unauthorized users.	Failed
UBTU-24-700040	SI-11 b	Ubuntu 24.04 LTS must be configured so that the "journalctl" command is owned by "root".	Passed

Control ID	NIST 800-53	Title	Status
UBTU-24-700050	SI-11 b	Ubuntu 24.04 LTS must be configured so that the "journalctl" command is group-owned by "root".	Passed
UBTU-24-700060	SI-11 b	Ubuntu 24.04 LTS must configure the directories used by the system journal to be group-owned by "systemd-journal".	Passed
UBTU-24-700070	SI-11 b	Ubuntu 24.04 LTS must configure the files used by the system journal to be group-owned by "systemd-journal".	Failed
UBTU-24-700080	SI-11 b	Ubuntu 24.04 LTS must configure the directories used by the system journal to be owned by "root".	Passed
UBTU-24-700090	SI-11 b	Ubuntu 24.04 LTS must configure the files used by the system journal to be owned by "root".	Passed
UBTU-24-700100	SI-11 b	Ubuntu 24.04 LTS must configure the /var/log directory to be group-owned by syslog.	Passed
UBTU-24-700110	SI-11 b	Ubuntu 24.04 LTS must configure the /var/log directory to be owned by root.	Passed
UBTU-24-700120	SI-11 b	Ubuntu 24.04 LTS must configure the /var/log directory to have mode "0755" or less permissive.	Failed
UBTU-24-700130	SI-11 b	Ubuntu 24.04 LTS must configure the /var/log/syslog file to be group-owned by adm.	Passed
UBTU-24-700140	SI-11 b	Ubuntu 24.04 LTS must configure /var/log/syslog file to be owned by syslog.	Passed
UBTU-24-700150	SI-11 b	Ubuntu 24.04 LTS must configure /var/log/syslog file with mode "0640" or less permissive.	Passed
UBTU-24-700300	SI-16	Ubuntu 24.04 LTS must implement nonexecutable data to protect its memory from unauthorized code execution.	Passed
UBTU-24-700310	SI-16	Ubuntu 24.04 LTS must implement address space layout randomization to protect its memory from unauthorized code execution.	Passed
UBTU-24-700320	SI-2 (6)	Ubuntu 24.04 LTS must be configured so that Advance Package Tool (APT) removes all software components after updated versions have been installed.	Failed
UBTU-24-700400	CM-6 b	Ubuntu 24.04 LTS must be a vendor-supported release.	Not Reviewed
UBTU-24-900040	AU-12 b	Ubuntu 24.04 LTS must be configured so that audit configuration files are not write-accessible by unauthorized users.	Failed
UBTU-24-900050	AU-12 b	Ubuntu 24.04 LTS must permit only authorized accounts to own the audit configuration files.	Passed
UBTU-24-900060	AU-12 b	Ubuntu 24.04 LTS must permit only authorized groups to own the audit configuration files.	Passed
UBTU-24-900070	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the su command.	Passed
UBTU-24-900080	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chfn command.	Passed
UBTU-24-900090	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the mount command.	Failed
UBTU-24-900100	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the umount command.	Failed
UBTU-24-900110	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the ssh-agent command.	Passed
UBTU-24-900120	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the ssh-keygen command.	Failed

Control ID	NIST 800-53	Title	Status
UBTU-24-900130	AU-12 c	Ubuntu 24.04 LTS must generate audit records for any use of the setxattr, fsetxattr, lsetxattr, removexattr, fremovexattr, and lremovexattr system calls.	Passed
UBTU-24-900140	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chown, fchown, fchownat, and lchown system calls.	Passed
UBTU-24-900150	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chmod, fchmod, and fchmodat system calls.	Passed
UBTU-24-900160	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the creat, open, openat, open_by_handle_at, truncate, and ftruncate system calls.	Failed
UBTU-24-900170	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the sudo command.	Passed
UBTU-24-900180	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the sudoedit command.	Failed
UBTU-24-900190	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chsh command.	Passed
UBTU-24-900200	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the newgrp command.	Passed
UBTU-24-900210	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chcon command.	Failed
UBTU-24-900220	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the apparmor_parser command.	Failed
UBTU-24-900230	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the setfacl command.	Failed
UBTU-24-900240	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chacl command.	Failed
UBTU-24-900250	AU-12 c	Ubuntu 24.04 LTS must generate audit records for the use and modification of faillog file.	Passed
UBTU-24-900260	AU-12 c	Ubuntu 24.04 LTS must generate audit records for the use and modification of the lastlog file.	Passed
UBTU-24-900270	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the passwd command.	Passed
UBTU-24-900280	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the unix_update command.	Failed
UBTU-24-900290	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the gpasswd command.	Passed
UBTU-24-900300	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the chage command.	Passed
UBTU-24-900310	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the usermod command.	Failed
UBTU-24-900320	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the crontab command.	Passed
UBTU-24-900330	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the pam_timestamp_check command.	Failed
UBTU-24-900340	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the init_module and finit_module syscalls.	Failed
UBTU-24-900350	AU-12 c	Ubuntu 24.04 LTS must generate audit records for successful/unsuccessful uses of the delete_module syscall.	Failed
UBTU-24-900510	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to modify the /etc/sudoers file occur.	Passed

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UBTU-24-900520	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to modify the /etc/sudoers.d directory occur.	Failed
UBTU-24-900540	AU-12 c	Ubuntu 24.04 LTS must generate audit records for any successful/unsuccessful use of unlink, unlinkat, rename, renameat, and rmdir system calls.	Failed
UBTU-24-900590	AU-12 c	Ubuntu 24.04 LTS must generate audit records for the /var/log/wtmp file.	Passed
UBTU-24-900600	AU-12 c	Ubuntu 24.04 LTS must generate audit records for the /var/run/utmp file.	Passed
UBTU-24-900610	AU-12 c	Ubuntu 24.04 LTS must generate audit records for the /var/log/btmp file.	Passed
UBTU-24-900730	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to use modprobe command.	Passed
UBTU-24-900740	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to use the kmod command.	Failed
UBTU-24-900750	AU-12 c	Ubuntu 24.04 LTS must generate audit records when successful/unsuccessful attempts to use the fdisk command.	Failed
UBTU-24-900920	AU-4	Ubuntu 24.04 LTS must allocate audit record storage capacity to store at least one week's worth of audit records, when audit records are not immediately sent to a central audit record storage facility.	Passed
UBTU-24-900950	AU-4 (1)	Ubuntu 24.04 LTS must have a crontab script running weekly to offload audit events of standalone systems.	Failed
UBTU-24-900960	AU-5 (1)	Ubuntu 24.04 LTS must immediately notify the system administrator (SA) and information system security officer (ISSO) (at a minimum) when allocated audit record storage volume reaches 75 percent of the repository maximum audit record storage capacity.	Failed
UBTU-24-900980	AU-5 a	Ubuntu 24.04 LTS must alert the system administrator (SA) and information system security officer (ISSO) (at a minimum) in the event of an audit processing failure.	Passed
UBTU-24-901220	AU-8 b	Ubuntu 24.04 LTS must record time stamps for audit records that can be mapped to Coordinated Universal Time (UTC) or Greenwich Mean Time (GMT).	Passed
UBTU-24-901230	AU-9, AU-9 a	Ubuntu 24.04 LTS must configure audit tools with a mode of "0755" or less permissive.	Passed
UBTU-24-901240	AU-9, AU-9 a	Ubuntu 24.04 LTS must configure audit tools to be owned by root.	Passed
UBTU-24-901250	AU-9, AU-9 a	Ubuntu 24.04 LTS must configure the audit tools to be group owned by root.	Passed
UBTU-24-901260	AU-9	Ubuntu 24.04 LTS must have directories that contain system commands set to a mode of "0755" or less permissive.	Passed
UBTU-24-901270	AU-9	Ubuntu 24.04 LTS must have directories that contain system commands owned by root.	Passed
UBTU-24-901280	AU-9	Ubuntu 24.04 LTS must have directories that contain system commands group-owned by root.	Passed
UBTU-24-901300	AU-9 a, AU-9 a	Ubuntu 24.04 LTS must be configured so that audit log files are not read or write-accessible by unauthorized users.	Failed
UBTU-24-901310	AU-9 a, AU-9 a, AU-9 a	Ubuntu 24.04 LTS must be configured to permit only authorized users ownership of the audit log files.	Passed
UBTU-24-901350	AU-9 a, AU-9 a, AU-9 a	Ubuntu 24.04 LTS must permit only authorized groups ownership of the audit log files.	Failed
UBTU-24-901380	AU-9 a	Ubuntu 24.04 LTS must be configured so that the audit log directory is not write-accessible by unauthorized users.	Passed

Control ID	NIST 800-53	Title	Status
UBTU-24-909000	AU-9 a	Ubuntu 24.04 LTS audit system must protect auditing rules from unauthorized change.	Passed
UBTU-24-909890	AU-9 (3)	Ubuntu 24.04 LTS must use cryptographic mechanisms to protect the integrity of audit tools.	Failed

